



Original Articles

When having two names facilitates lexical selection: Similar results in the picture-word task from translation distractors in bilinguals and synonym distractors in monolinguals[☆]

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ABSTRACT

We report five experiments using the picture-word task to examine lexical selection by comparing the effects of translation distractors in bilinguals and synonym distractors in monolinguals. Three groups of bilinguals named objects in their L1 or L2, and English monolinguals named objects using common names (e.g., *DOG* = “dog”) or, in a novel manipulation, using synonymous alternative names (e.g., *DOG* = “hound”, *GLASSES* = “spectacles”). All studies produced strikingly similar results. When bilinguals named in L1, there was a small facilitation effect from translation distractors, but larger facilitation when they named in L2. When monolinguals produced common names, there was no reliable effect from synonym distractors, but facilitation when they produced alternative names. (There were also strong identity facilitation effects in all naming conditions.) We discuss the relevance of these results for the debate concerning the role of competition in lexical selection and propose that for speech production there are direct facilitatory connections between the lexical representations of translations in bilinguals (and between synonyms in monolinguals). The effects of synonyms in monolinguals appear to “simulate” the effects found for translations in bilinguals, which suggest that there are commonalities in monolingual and bilingual lexical selection.

1. Introduction

A central component of speech production is the process of lexicalization, which refers to the cognitive operations whereby a semantic representation (e.g., <small mammal>, <pet>, <barks>, etc.) activates a word to be spoken (e.g., Caramazza, 1997; Levelt, Roelofs, & Meyer, 1999; Roelofs, 1992). It is generally accepted that the conceptual and semantic representations aroused by a communicative intent, or by a stimulus object to name, will activate a number of related words (e.g., ‘dog’, ‘hound’, ‘terrier’, ‘fox’, ‘cat’, etc.), if to varying degrees. Levelt et al. (1991) refer to the set of activated words as the “semantic cohort”. A major issue for theories of lexicalization is how people select a word from this cohort of activated candidates to produce in speech, and a continuing debate concerns the role of competition in the process of lexical selection (see Spalek, Damian, & Bölte, 2013).

Finkbeiner, Gollan, and Caramazza (2006) stated that “the closer two lexical representations are in meaning the more difficult it will be to select the correct one” (p. 153), and called this the “hard problem” of

lexical selection. This “hard problem” pertains to both within-language synonyms (e.g., couch–sofa, error–mistake) and bilingual translations. Finkbeiner et al. argued that, in bilingual speakers, “the hard problem is extensive because virtually every concept is associated with synonymous lexical nodes” (p. 153). Bilingual (and multilingual) speakers will always have two (or more) words to name the same object and, more generally, to express the same concept in speech. Their “hard problem” has indeed been hard to solve theoretically, although bilinguals generally appear not to find it too hard in practice. Our research aims to compare bilingual and monolingual lexical selection in the picture-word interference task by examining the effects of translation distractors in bilinguals and synonym distractors in monolinguals.

How are bilinguals able to “choose” the language of a word to produce? A number of logically possible solutions to the bilingual “hard problem” have been proposed. First, the message-level conceptual representation may be so effectively specifying that lexical representations in the non-target language would not be activated; this is the essence of La Heij’s (2005) “complex access, easy selection” approach to

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bilingual word production. Second, lexical representations in the non-target language may be activated but effectively ignored by the selection mechanism; this is the basis of the language-specific selection model championed by Costa, Miozzo, and Caramazza (1999) and Roelofs, Piai, and Garrido Rodriguez (2011). Third, the activation levels of lexical representations in the target language may be increased, as suggested by de Bot's (2004) "multilingual processing model". Fourth, lexical representations in the non-target language may be actively inhibited as part of the selection mechanism (Green, 1986, 1998). (Some accounts of monolingual lexical selection also propose the active inhibition of non-selected competitor words; Wheeldon & Monsell, 1994.) Finally, lexical representations in both languages may be activated, and speakers simply select the most activated word irrespective of its language (Finkbeiner et al., 2006).

All these possible solutions will need to be supplemented by an extra-lexical executive system to permit various aspects of language control. Such a control system is necessary to enable bilinguals to speak consistently in an intended language, to prevent the production of words in the non-intended language, and to allow flexibility in language choice (e.g., when translating words). The control system might operate at an input level, at an output level, or a combination of the two. At an input level, it could increment the activation of words in the intended or task-required language. At an output level, it could permit the production of words only in the intended language, perhaps as part of a more general "verbal self-monitoring" system of the kind proposed to repair slips and to prevent the production of nonwords (Dhooge & Hartsuiker, 2012; Hartsuiker & Kolk, 2001).

Lexical selection has often been studied using the picture-word interference task, in which participants are required to name a pictured object and to ignore a printed word superimposed upon it. Many studies have reported that naming times are slowed by distractor words from the same semantic category as the target (e.g., DOG + cat is slower than DOG + hat)¹ and this interference effect has been interpreted as reflecting a process of competition between the activated lexical representations of related words (e.g., Roelofs, 1997; Schriefers, Meyer, & Levelt, 1990; Starreveld & La Heij, 1996). However, studies have found facilitation from some semantically related words, such as the names of (unseen) parts of target objects (e.g., BRAIN + neurons; Costa, Alario, & Caramazza, 2005), different-category associates (e.g., NEEDLE + haystack; Sailor, Brooks, Bruening, Seiger-Gardner, & Guterman, 2009), and thematically related words (e.g., LION + Africa; de Zubicaray, Hansen, & McMahon, 2013). Mahon, Costa, Peterson, Vargas, and Caramazza (2007) found that related verbs (e.g., BED + sleep) produced facilitation and not interference, and they questioned the orthodox notion that lexical selection necessarily involves competition. They proposed the *response exclusion hypothesis* as a non-competitive account of performance in the picture-word task, and interpreted semantic interference effects in terms of a post-lexical process that excludes distractor words (see later). Our studies will be directly relevant to attempts to adjudicate between these theoretical alternatives concerning the role of competition in both monolingual and bilingual lexical selection.

Another result that is reliably observed in the picture-word task is the identity facilitation effect; naming times are faster when the distractor is the same name as the target object (e.g., DOG + dog is faster than DOG + key). In their seminal investigation of bilingual lexical selection, Costa et al. (1999) used a cross-language version of this identity effect to contrast two theoretical views. The *language-nonspecific* account proposes that lexical representations in both languages are considered by the selection mechanism (e.g., Poulish & Bongaerts, 1994).

In contrast, the *language-specific* account proposes that only the representations in the target language are considered as candidates by the selection mechanism. The experimental adjudication between these accounts centres on the effect of distractors that are translation-equivalents, such as GOS + perro (Catalan and Spanish, respectively, for 'dog'). If lexical selection were to be language-nonspecific, then translations, as particularly strong competitors, should interfere with naming. In stark contrast to this expectation, Costa et al. found that Catalan-Spanish bilinguals showed a facilitation effect from translation distractors, which supports language-specific selection. (Costa et al. referred to this as the "cross-language identity effect", but we shall call it the translation facilitation effect, as it is produced by a distractor word that is the translation of a target object name.) Although a simple view of the language-specific selection account might expect that translations should have no effect on lexical selection (as they are words in the not-to-be-produced language), Costa et al. interpreted the translation facilitation effect they found in terms of semantic priming from the translation distractor word to the target object that occurs prior to lexical selection.

Translation facilitation has been found by a number of studies (Costa, Colomé, Gómez, & Sebastián-Gallés, 2003; Roelofs, Piai, Garrido Rodriguez, & Chwilla, 2016; Roelofs et al., 2011), but it appears to be larger when bilinguals name in L2 than in L1, although this has not been compared directly as most studies have examined naming in one language only. For instance, Costa et al. (1999) tested naming only in L1 and found translation effects of between 10 and 31 ms over their various experiments, whereas Hermans (2004) and Roelofs et al. (2011), who tested naming only in L2 by Dutch-English bilinguals, found effects of 80–110 ms and 95 ms respectively. Costa and Caramazza (1999) found that translation facilitation was larger when English-Spanish bilinguals named in L2 than when Spanish-English bilinguals named in L1 (56 vs. 27 ms), although this between-group contrast was not analysed. It is therefore important to establish the reliability of any possible language asymmetry in the magnitude of the translation facilitation effect, and so our first experiments examined bilinguals' naming in both their L1 and L2 using within-participants designs. We tested three groups of bilinguals who differ in their relative proficiency of L2 compared to L1 to examine the generality of translation facilitation effects. The second motivation for our experiments with bilinguals was to provide a solid empirical base against which we can compare the results of similarly designed experiments with monolingual speakers when the distractor words are synonyms of the target name (e.g., GLASSES + spectacles). This allows the investigation of possible parallels in the processing of translation-equivalents in bilinguals and close synonyms in monolinguals, as Finkbeiner et al.'s "hard problem" of lexical selection applies to both.

1.1. Effects of translation distractors in bilinguals

We report three experiments, each with a similar design. In separate blocks of trials, participants named objects in their L1 and their L2. Within each block, there were four distractor words for each object: the object's name in L1; an unrelated control word in L1; the object's name in L2; and an unrelated control word in L2 (e.g., for the picture DOG, the words were perro, hombro, dog, and lid). (Each stimulus object appeared four times in each language block, and so appeared eight times in the whole experiment.) For each naming language, we will compare the effects of distractors that are identical to the target name (e.g., PERRO + perro and DOG + dog) and those that were translations of the target name (e.g., PERRO + dog and DOG + perro), both assessed against unrelated control words. We expect that identical distractor words will produce a large congruency facilitation effect when naming in L1 and in L2. The theoretically important question is whether the effect from translation distractors will be larger when naming in L2 than in L1. Therefore, the effects from identical words and from translations will be analysed separately.

¹ We use the notational conventions of expressing target object names in upper case, underlining printed distractor words (e.g., DOG + hound), using 'single quotes' for lexical representations (e.g., 'dog', 'perro'), and using "double quotes" for spoken responses (e.g., "dog").

2. Experiment 1: Lexical facilitation from translations in Spanish-English bilinguals

2.1. Method

2.1.1. Participants

Sixteen Spanish-English bilinguals (10 women and six men) volunteered. The mean age was 29.4 years (ranging from 19 to 37 years). All participants were from a Spanish speaking country, but had lived and studied in England for at least one year (mean = 2.3 years); many were masters-level students. Their level of English proficiency, as indicated by the reading component of the Dialang English language assessment, was quite high; all 16 participants agreed with the statement “I can understand any correspondence with an occasional use of dictionary”, 15/16 agreed with “I can understand in detail a wide range of long, complex texts provided I can re-read difficult sections”, and 12/16 agreed with “I can understand and interpret practically all forms of written language”. All participants had normal or corrected-to-normal vision, and none reported having any reading difficulties in either language.

2.1.2. Stimulus materials

Twenty-two pictures were selected from Snodgrass and Vanderwart (1980) that had 100% name agreement in British English (Barry, Morrison, & Ellis, 1997) and whose Spanish and English names were not cognates. Each picture was paired with four distractor words: its Spanish name, an unrelated Spanish control, its English name, and an unrelated English control. The English names and their controls were matched on English word frequency (Brysbaert & New, 2009) (74.6 and 73.9 per million, $t < 0.2$), number of letters (4.64 and 4.64, $t = 0$), and number of syllables (1.32 and 1.32, $t = 0$). The Spanish names and their controls were matched on Spanish word frequency (Pérez & Navalón, 2005) (65.8 and 59.1 per million, $t < 0.3$), number of letters (5.68 and 5.90, $t < 0.6$), and number of syllables (2.55 vs. 2.50, $t < 0.3$). All words were concrete nouns. There were 8 practice trials in each language. Appendix A lists the experimental stimuli used.

2.1.3. Procedure

The participants were tested individually on an Apple iMac computer using the software program SuperLab. Participants were told that a picture of an object, with a printed word superimposed on it, would appear in the centre of the computer screen, and were asked to name the picture (as quickly and as accurately as possible) whilst trying to ignore the printed word. The distractor words were presented in lower-case Verdana font size 36 and appeared on the middle of the pictures. The picture and word were presented simultaneously, and both remained on the screen until the participant's naming response was detected by the computer's built-in microphone connected to SuperLab's voice-key. The experimenter then classified each response as being correct or incorrect, and the next trial began 1 s later. The pictures and words, when presented on the computer screen, were between 8×5 cm and 4×3 cm in size. The participants completed two blocks of trials, one naming in Spanish and one in English, with the order of naming language counterbalanced. Before each block, the participants were shown all the pictures, together with their names in the target language. Following this familiarization, they received 8 practice trials and then the 88 experimental trials (22 pictures $\times$ 4 distractors) presented in a randomized order. The sequence of familiarization, practice and experimental trials was then repeated for the other target language.

2.2. Results

Responses that were incorrect, hesitant or failed to stop the timer were excluded, as were responses below 250 ms or above 2.5 s. Overall, 3.3% of responses were excluded and these were not analysed statistically. Harmonic means of non-excluded responses were calculated, for

Table 1

Results from Spanish-English bilinguals in Experiment 1. Mean Object Naming Latencies (in Milliseconds), Standard Deviations, and Rates of Excluded trials (%Ex) by Naming Language and Type of Distractor.

Distractor type	Naming in L1 Spanish			Naming in L2 English		
	Mean	SD	%Ex	Mean	SD	%Ex
(a) Within-Language (identity)	PERRO + <u>perro/hombro</u>			DOG + <u>dog/key</u>		
Identity	707	102	4.6	713	52	1.7
Unrelated	807	114	4.5	807	73	1.7
Facilitation =	100			94		
(b) Cross-language (translation)	PERRO + <u>dog/key</u>			DOG + <u>perro/hombro</u>		
Translation	791	97	4.0	798	63	3.4
Unrelated	796	113	4.3	836	75	2.3
Facilitation =	5			38		

Note: Facilitation = Unrelated minus Identity or Translation.

each participant and for each item, in each condition. Harmonic means (rather than medians, or means after variously defined outliers were excluded) were calculated as the reciprocal transformation they involve has the effect of reducing the skew of naming latency distributions towards slower responses (see Ratcliff, 1993). In studies of object naming, harmonic means of naming times have been used less often than arithmetic means after outliers have been removed. In this, and in all subsequent experiments, similar analyses were also conducted on trimmed arithmetic means of naming latencies in each condition after excluding naming times that were 2.5 standard deviations above (or below) each participant's mean in each naming task. These analyses showed the same patterns as reported in the text, and the means for each experiment are shown in Appendix F.

The means of harmonic mean naming times in each condition are shown in Table 1. There were large facilitation effects from identical distractors both for naming in L1 and L2. In contrast, the facilitation effect from translation distractors (which was smaller overall) was larger for naming in L2 than in L1.

The data were analysed separately for the effects of within-language (identity) and cross-language (translation) distractors by 2×2 related ANOVAs (both by participants and items), with the factors of naming language (L1 vs. L2) and distractor relationship (related vs. unrelated controls). Both factors were manipulated within participants and within items.

(a) *Identity*. There was no main effect of naming language, $F_1 = 0.02$, $F_2 = 0.93$; naming times in Spanish were only 3 ms faster than in English. There was a large main effect of relationship, $F_1(1, 15) = 92.4$, $MSe = 1628$, $p < 0.0001$, $F_2(1, 21) = 115.8$, $MSe = 1812$, $p < 0.0001$; naming times were 97 ms faster with identical than unrelated distractors. The interaction between the two factors was absent, $F_1 = 0.2$, $F_2 = 0.3$.

(b) *Translation*. The main effect of naming language was not significant in the analysis by participants, $F_1(1, 15) = 1.1$, but was in the analysis by items, $F_2(1, 21) = 8.8$, $MSe = 2429$, $p < 0.01$; overall, naming times in Spanish were 24 ms faster than in English. There was a main effect of relationship, $F_1(1, 15) = 9.4$, $MSe = 782$, $p < 0.01$, $F_2(1, 21) = 3.1$, $MSe = 2440$, $p < 0.1$, and an interaction between the two factors that was significant by participants, $F_1(1, 15) = 7.8$, $MSe = 519$, $p < 0.01$, but not quite by items, $F_2(1, 21) = 3.4$, $MSe = 2333$, $p < 0.1$. The translation facilitation effect was significant for naming in L2, $t_1(15) = 3.8$, $p < 0.01$, $t_2(21) = 2.3$, $p < 0.05$, but not for naming in L1, $t_1(15) = 0.6$, $t_2(21) = 0.6$.

2.3. Discussion

Identical distractors (e.g., PERRO + perro and DOG + dog) produced a substantial facilitation effect of equivalent magnitude when naming in

L1 Spanish and L2 English. Translation distractors produced a reliable facilitation effect when naming in L2 (e.g., DOG + perro), but not when naming in L1 (e.g., PERRO + dog), where the small effect (of 5 ms) was not reliable.

As we found no interference at all from translations, these results support the language-specific account of bilingual lexical selection (Costa et al., 1999), but they also show that translations are more facilitating of naming in L2 than in L1. Similar asymmetric effects in bilinguals have been found in other situations; for example, Gollan, Forster, and Frost (1997) found that masked translation priming of visual lexical decision in Hebrew-English bilinguals was larger from L1-to-L2 than from L2-to-L1. Our demonstration of language asymmetry in the translation facilitation effect, in a within-subject comparison, is an important result that has significance for theoretical accounts of the organization of bilingual lexical representations for speech production. However, before considering such accounts, we shall report two replications to test the generality of the language asymmetry in the translation effect in different groups of bilinguals.

3. Experiment 2: Lexical facilitation from translations in Swedish-English bilinguals

This experiment tested a group of native Swedish speakers living in Sweden who have high levels of L2 English proficiency and may be considered to be more “balanced” bilinguals than those tested in Experiment 1. The experiment adopted the same design as the previous study, with one procedural difference: the distractor words used in the unrelated conditions were created by re-ordering the words in the related conditions rather than by constructing matched words (as in Experiment 1). This re-ordering procedure of distractor words is commonly used in monolingual studies using the picture-word task (e.g., Damian & Spalek, 2014).

3.1. Method

3.1.1. Participants

Thirty Swedish-English bilinguals (20 women and 10 men) volunteered. The mean age was 26.9 years (ranging from 18 to 48 years). All participants were native Swedish speakers, and rated their English proficiency as being high. Using a rating scale from 1 to 10, their overall English proficiency was 7.6; reading = 8.2; writing = 7.2; listening = 8.2; speaking = 7.5.

3.1.2. Stimulus materials

Forty-two pictures with non-cognate names in Swedish and English were taken from Snodgrass and Vanderwart (1980). The pictures had high name agreement in British English (mean = 98%; Barry et al., 1997) and in Swedish (mean = 96%, based on a separate pilot study with 38 native Swedish speakers). The mean frequency of the Swedish words was 28.1 per million (Kokkinakis & Volodina, 2011), and the mean frequency of the English words was 43.2 per million (Brysbaert & New, 2009); these did not differ reliably, $p < 0.25$. In each naming language, each object was presented four times, once each with the four types of distractor word. There were 10 practice trials in each language. Appendix B lists the experimental stimuli used.

3.1.3. Procedure

The participants were tested individually on an Apple laptop computer running SuperLab, with the same task instructions, stimulus presentation and procedure as used in Experiment 1. Participants completed two blocks of trials, one naming in Swedish and one in English, with the order of naming language counterbalanced, and, before each block, were familiarized with the pictures and their target names. Each block had 8 practice trials before the 168 experimental trials (42 pictures $\times$ 4 distractors), which were presented in a randomized order.

Table 2

Results from Swedish-English bilinguals in Experiment 2. Mean Object Naming Latencies (in Milliseconds), Standard Deviations, and Rates of Excluded trials (%Ex) by Naming Language and Type of Distractor.

Distractor type	Naming in L1 Swedish			Naming in L2 English		
	Mean	SD	%Ex	Mean	SD	%Ex
(a) Within-Language (identity)	STOL + <u>stol/kanin</u>			CHAIR + <u>chair/dress</u>		
Identity	721	91	3.1	718	81	2.9
Unrelated	890	119	6.3	883	94	6.3
Facilitation =	179			165		
(b) Cross-language (translation)	STOL + <u>chair/dress</u>			CHAIR + <u>stol/kanin</u>		
Translation	832	98	5.2	760	74	4.5
Unrelated	884	105	6.7	888	117	7.0
Facilitation =	52			128		

Note: Facilitation = Unrelated minus Identity or Translation.

3.2. Results and discussion

Responses that were incorrect, hesitant or failed to stop the timer were excluded, as were responses below 250 ms or above 2.5 s. Overall, 5.2% of responses were excluded. Harmonic means of non-excluded responses were calculated, for each participant and for each item, in each condition. Table 2 presents the results, which show that there was a substantial identity facilitation effect for naming in both L1 and L2, and a translation facilitation effect that is larger when naming in L2 than in L1. This pattern of results is similar to that observed in Experiment 1, although the translation effects were larger and observed when naming both in L1 and L2. Separate 2×2 related ANOVAs were conducted for the within-language (identity) and cross-language (translation) distractors, both by participants and by items, with the factors of naming language and relationship.

(a) *Identity*. The main effect of naming language was absent, $F_1 = 0.04$, $F_2 = 0.01$; overall, naming times in Swedish were 5 ms slower than in English. The main effect of relationship was large, $F_1(1, 29) = 253.2$, $MSe = 3325$, $p < 0.0001$, $F_2(1, 41) = 485.5$, $MSe = 2469$, $p < 0.0001$; naming times were 173 ms faster with identical than unrelated distractors. There was no interaction between the two factors, $F_1 = 0.08$, $F_2 = 0.01$.

(b) *Translation*. The main effect of naming language was not significant in the analysis by participants, $F_1(1, 29) = 1.9$, $MSe = 17819$, $p < 0.2$, but was in the analysis by items, $F_2(1, 41) = 11.4$, $MSe = 4197$, $p < 0.005$; overall, naming times were 59 ms slower in Swedish than in English. The main effect of relationship was significant, $F_1(1, 29) = 117.8$, $MSe = 2082$, $p < 0.0001$, $F_2(1, 41) = 102.4$, $MSe = 3089$, $p < 0.0001$; overall, naming times were 90 ms faster with translation than unrelated distractors. The important interaction between the two factors was also significant, $F_1(1, 29) = 29.5$, $MSe = 1447$, $p < 0.0001$, $F_2(1, 41) = 19.8$, $MSe = 2780$, $p < 0.0001$; the translation facilitation effect was larger for naming in L2 than L1 (128 vs. 52 ms), but was reliable for both naming in L1, $t_1(29) = 4.5$, $p < 0.01$, $t_2(41) = 4.2$, $p < 0.01$, and in L2, $t_1(29) = 10.9$, $p < 0.01$, $t_2(43) = 10.1$, $p < 0.01$.

Discussion. The results from a group of balanced Swedish-English bilinguals showed a substantial identity facilitation effect for naming in both L1 and L2, and a translation facilitation effect that is larger when naming in L2 than in L1. We therefore replicated the language asymmetry in the translation facilitation effect found in Experiment 1, and showed that it generalizes to bilinguals with high levels of L2 proficiency.

4. Experiment 3: Lexical facilitation from translations in English-French bilinguals

The bilinguals tested in our first two experiments had English as their L2. In order to exclude the possibility, however unlikely, that our results are limited to peculiarities of naming in English as an L2, Experiment 3 tested a group of native English speakers who have lived in France for at least 10 years and have high levels of L2 French proficiency. They may be considered to be balanced bilinguals, and all reported using both French and English on a daily basis.

4.1. Method

4.1.1. Participants

Twenty English-French bilinguals (12 women and eight men) living in France were tested. They were aged between 20 and 65, with a mean age of 41. Six participants were English teachers in France, and the others included family members and acquaintances.

4.1.2. Stimulus materials

Thirty pictures with common names were selected from Snodgrass and Vanderwart (1980) that had at least 90% (mean = 96.2%) name agreement in British English (Barry et al., 1997) and whose English and French names were not cognates. Each picture was paired with four distractor words: its English name, an unrelated English control, its French name, and an unrelated French control. The mean frequency of the English words was 96.2 per million (Brysbaert & New, 2009) and the mean frequency of the French words was 103.5 per million (New, Pallier, Brysbaert, & Ferrand, 2004); these did not differ, $p < 0.8$. The distractor words were presented in lower-case Geneva font size 36 and appeared in the middle of the pictures. Appendix C lists the experimental stimuli used.

4.1.3. Procedure

The participants were tested individually on an Apple laptop computer running SuperLab, with the same task instructions, stimulus presentation and procedure as used in Experiment 1. Participants completed two blocks of trials, one naming in English and one in French, with the order of naming language counterbalanced, and, before each block, were familiarized with the pictures and their target names. Each block had 8 practice trials before the 120 experimental trials (30 pictures $\times$ 4 distractors), which were presented in a randomized order. The experimenter then classified each response as being correct or incorrect, and the next trial began 250 ms later.

4.2. Results and discussion

Responses that were incorrect, hesitant or failed to stop the timer were excluded, as were responses below 250 ms or above 2.5 s. Overall, 3.8% of responses were excluded. Harmonic means of non-excluded responses were calculated, for each participant and for each item, in each condition. Table 3 presents the results, which show a very similar pattern to that found in Experiment 2; the translation facilitation effect was larger when naming in L2 than in L1. Separate 2×2 related ANOVAs were conducted for the identity and translation distractors, both by participants and by items, with the factors of naming language and relationship.

(a) *Identity*. The main effect of naming language was not significant by participants, $F_1(1, 19) = 1.3$, but was by items, $F_2(1, 29) = 11.5$, $MSe = 1085$, $p < 0.01$; overall, naming times in English were 16 ms faster than in French. The main effect of relationship was very large, $F_1(1, 19) = 202.7$, $MSe = 1466$, $p < 0.0001$, $F_2(1, 29) = 278.5$, $MSe = 1603$, $p < 0.0001$; naming times were 122 ms faster with

Table 3

Results from English-French bilinguals in Experiment 3. Mean Object Naming Latencies (in Milliseconds), Standard Deviations, and Rates of Excluded trials (%Ex) by Naming Language and Type of Distractor.

Distractor type	Naming in L1 English			Naming in L2 French		
	Mean	SD	%Ex	Mean	SD	%Ex
(a) <i>Within-Language (identity)</i>	<i>DOG + dog/bed</i>			<i>CHIEN + chien/lit</i>		
Identity	754	97	1.8	760	89	1.7
Unrelated	864	110	3.3	894	104	8.6
Facilitation =	110			134		
(b) <i>Cross-Language (translation)</i>	<i>DOG + chien/lit</i>			<i>CHIEN + dog/bed</i>		
Translation	801	74	3.8	821	87	4.7
Unrelated	843	92	2.5	897	105	4.3
Facilitation =	42			76		

Note: Facilitation = Unrelated minus Identity or Translation.

identical than unrelated distractors. The interaction between the two factors was not significant by participants, $F_1(1, 19) = 2.1$, $p < 0.2$, but just reached significance by items, $F_2(1, 29) = 4.7$, $MSe = 808$, $p < 0.05$; as can be seen in Table 2, the identity facilitation effect was slightly larger when naming in L2 than in L1 (134 vs. 110 ms), but it was significant for both ($t > 9.0$).

(b) *Translation*. There were main effects of both naming language, $F_1(1, 19) = 6.1$, $MSe = 4408$, $p < 0.05$, $F_2(1, 29) = 12.3$, $MSe = 2962$, $p < 0.001$, and relationship, $F_1(1, 19) = 45.3$, $MSe = 1536$, $p < 0.0001$, $F_2(1, 29) = 51.2$, $MSe = 1754$, $p < 0.0001$; overall, naming times were 37 ms faster in L1 than L2, and 59 ms faster with translation than unrelated distractors. Importantly, however, the interaction between the two factors was also significant, $F_1(1, 19) = 10.1$, $MSe = 570$, $p < 0.01$, $F_2(1, 29) = 6.2$, $MSe = 1697$, $p < 0.05$; the translation facilitation effect was larger for naming in L2 than L1 (76 vs. 42 ms), although it was reliable for both L1 and L2, $t > 3.3$, $p < 0.01$.

4.2.1. Discussion

The results of Experiment 3 from a group of balanced English-French bilinguals, immersed in *la vie française*, replicated the asymmetry of translation facilitation, which was larger when naming in L2 than in L1.

4.3. Discussion of experiments with bilinguals

All bilinguals showed larger translation facilitation when naming in L2 than in L1. The translation facilitation effect was significant for naming in L2 in all three experiments, and was significant for naming in L1 in Experiments 2 and 3. The translation facilitation effects were larger for the Swedish-English and English-French bilinguals, who had high levels of L2 proficiency, than for the less proficient Spanish-English bilinguals.

If translation facilitation is due solely to semantic priming, as argued by Costa et al. (1999), then it is not clear why there should be the language asymmetry we observed. Semantic interference effects in the picture-word task have been found to be essentially equal from within- and cross-language category-related distractor words in bilinguals both when naming in L1 (e.g., Costa et al., 1999) and in L2 (e.g., Smith & Kirsner, 1982). This suggests that semantic effects are the same for naming in L2 and in L1, and yet our three experiments show that the translation facilitation effect was reliably larger when naming in L2 than in L1.

The significant facilitation from translations when naming in L1

(e.g., SÄNG + bed, BED + lit) in the bilinguals in Experiments 2 and 3 may have arisen from a number of sources. It is possible that reading processes for L2 words are more efficient in bilinguals with greater L2 proficiency, which would produce greater word-to-target semantic priming, but [Piai, Roelofs, and Roete \(2015\)](#) found for Dutch participants that within-language semantic interference effects in the picture-word task were not associated with level of reading ability.

To advance an integrated theoretical account of our results, we propose that the lexical representations of translations are interconnected in a mutually facilitatory but asymmetrical fashion, such that more activation is passed from L1 to L2 translations than vice versa. For all bilinguals, dominant L1 lexical representations always assist L2 representations of their translations; thus, L1-to-L2 activation flow contributes to the translation facilitation of naming in L2. As L2 proficiency increases, the activation flow from L2-to-L1 connections will also increase. This reverberating activation flow between the bidirectional connections of translations will increase the magnitude of the translation facilitation effect when naming in both L1 and L2 (which is exactly the pattern we found in Experiments 2 and 3).

4.3.1. Phonological activation of translations

Our findings clearly demonstrate that translation distractors facilitate and never interfere with naming times, and therefore support the language-specific (or within-language) rather than the language non-specific theory of bilingual lexical selection. The language-specific view accepts that both languages will be activated (as stimulus objects are assumed to activate names in both languages), but proposes that only words in the target language “are considered for selection” ([Costa et al., 1999, p. 397](#)) or “compete for selection” ([Roelofs et al., 2016, p. 2](#)). However, [Hermans \(2004\)](#) has questioned the language-specific account on the basis of results concerning cross-language phonological effects in the picture-word task. First, [Hermans, Bongaerts, De Bot, and Schreuder \(1998\)](#) found that when Dutch-English bilinguals named in L2, auditory distractor L1 words that were phonologically related to the object’s L1 (i.e., non-target) name produced an interference effect on naming. For example, MOUNTAIN + “berm” was slower than an unrelated control condition because *berm* is phonologically related to “berg” (the L1 translation of the target “mountain”), and so increases its status as a lexical competitor to the target. This effect was replicated in balanced Spanish-Catalan bilinguals naming in L2 ([Costa et al., 2003](#)). Although the “phono-translation” effect may be taken as support for a language non-specific account of bilingual lexical selection, [Costa et al. \(2003\)](#) argue that it is due to cross-language influence at the post-lexical level of phonological encoding of a language-specific selected naming response. This argument assumes that the process of lexical selection itself operates entirely in a language-specific manner, but that the subsequent phonological encoding of selected names may be affected by words from any language.

Second, [Hermans \(2004\)](#) found that when Dutch-English bilinguals named in L2, printed L1 distractor words whose L2 translations are phonologically related to the target name produce a facilitation effect. For example, MONKEY + geld was faster than an unrelated control condition because the translation of geld (“money”) is phonologically related to the target name “monkey”. This effect (of 35 ms) was observed only when the distractor words were also translations of the names of the pictures used in the experiment; it was not reliable when compared to unrelated words not in the set of possible names of the stimulus objects. However, [Costa et al. \(1999\)](#) found that naming in L1 showed no effect of either L1 or L2 distractor words whose translations were phonologically related to the target.

The “phono-translation” results of [Hermans et al. \(1998\)](#) show that distractor words that are phonologically related to translations interfere with naming-times, whereas the results of our three experiments show that the translations themselves presented as distractor words facilitate naming-times. For example, naming a picture “mountain” would be slowed by the indirect, phonologically mediated impact of the

distractor word berm, via its translation ‘berg’, yet would be speeded by the lexically direct impact of the distractor word berg itself. This conjunction of results is paradoxical. At present, it is difficult to decide how critical the phonologically mediated results are for adjudicating between language-specific and language non-specific accounts of bilingual lexical selection, as opposed to revealing language non-specific post-selection phonological processing. However, our results clearly suggest that the lexical processes that produce facilitation out-weigh any possible effect of phonologically mediated interference.

4.3.2. Effects of order of naming in L1 and L2

Some studies of bilinguals have reported an asymmetry in the “after-effects” ([Branzi, Martin, Abutaleb, & Costa, 2014](#)) from first naming in L1 and then naming the same objects in L2, compared to first naming in L2 and then in L1. [Misra, Guo, Bobb, and Kroll \(2012\)](#) found that Chinese-English bilinguals showed significantly faster naming in L2 after the same objects had been first named in L1, whereas naming in L1 was actually slightly slower after having previously named the same objects in L2. [Branzi et al. \(2014\)](#) also found that L2 naming was facilitated from previous naming in L1, whereas L1 naming showed no reliable effect from previous naming in L2. [Van Assche, Duyck, and Gollan \(2013\)](#) found that two groups of bilinguals both showed reduced L1 performance in a verbal fluency task after first producing exemplars in L2 from the same categories, but L2 fluency was not influenced by prior production of L1 words from the same categories. The asymmetric after-effects observed in these experiments have been taken as evidence of persistent inhibition of L1 during naming in L2.

In each of our three experiments reported here, half of the bilingual participants named the objects in L1 and then named the same objects in L2, whereas half had the opposite order of presentation of blocks of trials. In order to examine effects of order of language production, we combined the data from all three experiments in order to achieve greater power. The results are shown in [Table 4](#), and were analysed by two mixed 3-way ANOVAs (by participants only), separately for identity and translation distractors. These 2 × 2 × 2 ANOVAs had the between-participant factor of order of language production (L1 first vs. L2 first) and the within-participant factors of naming language and distractor relationship.

(a) *Identity*. The analysis repeated effects already reported for each experiment: there was a significant main effect of distractor type, $F_1(1, 64) = 371.9, p < 0.0001$, but no main effect of language, $F_1 < 1$, and no language by distractor interaction, $F_1 < 1$.

The main effect of order of language production was significant, $F_1(1, 64) = 3.7, p < 0.05$. Bilinguals who produced L1 names first were overall 38 ms faster than those who produced L2 names first. The interaction between order of production and language was significant,

Table 4
Effects of Order of Language Production in Bilinguals. Mean Object Naming Latencies (in Milliseconds) in each condition collapsed over all bilinguals tested in Experiments 1, 2 and 3.

Naming language =	Named in L1 first		Named in L2 first	
	L1	L2	L1	L2
(a) Within-Language (identity)				
Identity	703	720	753	739
Unrelated	815	873	909	864
Facilitation =	112	153	157	125
(b) Cross-language (translation)				
Translation	782	784	843	792
Unrelated	810	877	891	879
Facilitation =	28	93	48	87
Mean naming time	778	814	849	819

Note: Facilitation = Unrelated minus Identity or Translation. Mean naming time = mean of each naming language over all four distractor types.

$F_1(1, 64) = 16.3, p < 0.01$, which showed that the bilinguals who named first in L1 were 35 ms faster in L1 than in L2, whereas those who named first in L2 were 29 ms slower in L1 than in L2. In other words, both groups showed equally slower naming in their second block of naming trials, irrespective of language. The interaction between order and distractor type was absent, $F_1 < 1$; the identity facilitation effect was essentially the same for bilinguals who first named in L1 (132 ms) or first named in L2 (141 ms).

(b) *Translation*. This analysis also repeated effects already reported: there was a significant main effect of distractor type, $F_1(1, 64) = 114.4, p < 0.0001$, no main effect of language, $F_1 < 1$, and a significant interaction between language and distractor type, $F_1(1, 64) = 44.1, p < 0.001$; the translation facilitation effect was larger for naming in L2 than L1.

There was a significant main effect of order of language production, $F_1(1, 64) = 3.8, p < 0.05$; bilinguals who produced L1 names first were 37 ms faster than those who produced L2 names first. The significant interaction between order and language, $F_1(1, 64) = 14.0, p < 0.001$, showed that bilinguals who named first in L1 were 35 ms faster in L1 than in L2, whereas those who named first in L2 were 31 ms slower in L1 than in L2. The interaction between order and distractor type was absent, $F_1 < 1$, showing that the translation facilitation effect was essentially the same for bilinguals who first named in L1 (59 ms) or first named in L2 (67 ms).

4.3.3. Discussion

These analyses show two main results concerning the effects of order of language production. First, we did not replicate the order effects found in previous studies. Indeed, we found that the overall naming times in L1 and L2 showed symmetric and negative after-effects from naming previously in the other language. Bilinguals were equally slower naming in L2 after having named the same objects in L1, and slower naming in L1 after previous naming in L2. In other words, they were equally slower when naming in the second block of trials. It is unclear what may be concluded from this finding concerning the debate as to whether the results of previous studies (e.g., Misra et al., 2012) can be interpreted in terms of inhibitory processes applied to L1 when naming in L2 or of possible “over-activation” of L2 from greater repetition priming. Our experiments using the picture-word task were not directly appropriate to informing this debate: the pictures to be named were presented with both same- and different-language distractor words, were repeated four times in each naming block, and their names in each target language were familiarized prior to the main experimental trials. Second, the important feature of these results for the purpose of the present studies is that the pattern of facilitation effects from identical and translation distractors did not differ reliably as a function of order of language production.

4.4. Effects of synonym distractors in monolinguals

Our next two experiments attempt to build a connection between our findings concerning the cross-language influences of translation distractors in bilingual lexical selection and possible within-language influences of synonym distractors in monolingual lexical selection. Translations are words in different languages that mean and refer to the same thing. Synonyms also mean and refer to essentially the same thing, but are words in the same language. Both translations and synonyms may be seen as *alternative names* for the same object, and, more generally, they are words that equally match (and can equally express in speech) the same semantic representation. Whereas translations and synonyms both face the same “hard problem” of lexical selection in language production, it is important to note that bilinguals face this problem for all lexicalized concepts, whereas monolinguals face it only for concepts that permit synonymous words.

In this section, we report two experiments examining the effects of within-language synonym distractors in monolingual English speakers to see if these related words produce similar results as translations do for bilinguals. The experiments examine naming when the related distractors were the same name (e.g., DOG + dog) or a synonym of the target name (e.g., DOG + hound, GLASSES + spectacles). We are calling these words “synonyms” because they are both good labels for the presented picture. However, not all the words were exactly synonymous in the sense of being interchangeable in every context. We discuss this issue later. The “hard problem” of lexical selection in bilinguals applies equally to within-language synonyms. Synonyms are pairs of words that equivalently (or very closely) match a semantic specification, although differences in their frequency of usage will affect their resting lexical activation levels. The effect of synonym distractors represents a critical test of the theory that monolingual lexical selection involves competition. Any parallels between the effects of synonyms in monolinguals and translations in bilinguals will reveal important commonalities in monolingual and bilingual lexical selection.

The novel procedure used in Experiments 4 and 5 was that participants were asked to name the same objects in two tasks: *common word* naming (e.g., naming as “dog” or as “glasses”) and *alternative synonymous* naming (e.g., naming as “hound” or as “spectacles”). In the synonymous naming task, participants were first shown the pictures along with the required target names and were asked to produce only these alternative words as naming responses. We reasoned that synonym naming might provide a broad within-language analog of when bilinguals name in L2, and so we therefore expect that synonym distractors will produce the same pattern of results in monolinguals as translations do in bilinguals.

5. Experiment 4: Lexical facilitation from synonyms in monolinguals

5.1. Participants

Thirty-two monolingual native speakers of English (25 women and 7 men) studying at Essex University volunteered. Their mean age was 20.3 years (between 18 and 36 years old).

5.2. Stimulus materials

Twenty-one line drawings were selected from a variety of sources. The pictures were chosen so that each had a high common name agreement but also permitted a close synonym name. A set of 46 possible synonym pairs (e.g., dog–hound, car–automobile) was presented to a group of 9 postgraduates to rate for their degree of synonymy on a 5-point scale, ranging from 1 = completely different to 5 = practically the same. Each of the 21 pictures selected for the experiment had a mean rating of above 4 (mean = 4.47, ranging from 4.1 to 5.0). The selected pictures had a mean name agreement of 90.8% (ranging from 70% to 100%).

In each naming task, the 21 pictures appeared four times, each time with a different distractor word: the target name (e.g., DOG + dog, HOUND + hound); an unrelated matched control word (e.g., DOG + eye, HOUND + torch); a synonym of the target name (e.g., DOG + hound, HOUND + dog); and an unrelated matched control word (e.g., DOG + torch, HOUND + eye). The common names and their controls, all of which were concrete nouns, were matched on word frequency (Brysbaert & New, 2009; 58.8 and 56.4 per million, $t < 0.6$), number of letters (5.05 and 5.05, $t = 0$), and number of syllables (1.57 and 1.52, $t = 1.0$). The synonyms and their controls were matched on frequency (13.5 and 13.7, $t < 0.6$), number of letters (6.33 and 6.33, $t = 0$), and number of syllables (1.90 and 1.86, $t < 0.5$). The common names had higher word frequencies than the synonym names (59 vs. 14 per

million). Four different pictures were used for the practice trials. Appendix D shows the full list of the experimental stimuli.

5.3. Procedure

Participants were tested individually with the same equipment and the same general instructions (e.g., to name pictures and to ignore superimposed words) as used previously. The participants completed two tasks: once naming the pictures with their “common” names, and once naming them with their “alternative” (lower-frequency synonym) names. The order of naming task was counterbalanced. Before each naming task, the participants were shown the pictures together with their target names, and were asked to produce only these names. Following this familiarization in each task, the participants were shown all the pictures again, without their names, and were asked to produce the names they had been instructed to use. Any non-target names produced were corrected. Participants experienced no particular difficulty when asked to produce the synonym names. In each naming task, participants were given 4 practice trials, followed by 84 experimental trials (21 pictures × 4 words) presented in a randomized order. Each picture and distractor word remained on the screen until the participant gave a response. The experimenter classified each response as being acceptable or not, and the next stimulus appeared 1 s later.

6. Results

Responses that were incorrect, lip-smacks, or failed to stop the timer were excluded, as were responses below 250 ms or above 2 s. Overall, 3.6% of responses were excluded and these were not analysed statistically. Although there were some mis-naming errors in the “as hound” task, most exclusions were hesitations or verbal stumbles (e.g., “oh, it’s a er ...”).

Harmonic means of non-excluded responses were calculated, both for each participant and for each item, in each condition. The means of these harmonic means (by participants) are shown in Table 5. There was a substantial identity facilitation effect in both the common word (“as dog”) and alternative (“as hound”) naming tasks, but synonym distractors facilitated naming only in alternative naming task.

Separate 2 × 2 related ANOVAs (both by participants and items) were conducted to examine the effects of identical and synonym distractors, in a manner similar to the results of identical and translation distractors in Experiments 1, 2 and 3. The ANOVAs contained the factors of naming task (“as dog” vs. “as hound”) and distractor relationship (related vs. unrelated). Both factors were manipulated within participants and within items.

(a) Identity. There were main effects of naming task, $F_1(1, 31)$

$= 3.2$, $MSe = 13727$, $p < 0.1$, $F_2(1, 20) = 6.8$, $MSe = 3766$, $p < 0.05$, and distractor relationship, $F_1(1, 31) = 61.8$, $MSe = 3375$, $p < 0.0001$, $F_2(1, 20) = 90.9$, $MSe = 1599$, $p < 0.0001$; participants were 37 ms faster to produce common names than alternative synonym names, and were 81 ms faster with identical than unrelated distractors. There was a marginal interaction between the two factors, $F_1(1, 31) = 3.44$, $MSe = 1298$, $p < 0.1$, $F_2(1, 20) = 3.61$, $MSe = 1238$, $p < 0.1$, reflecting that the identity facilitation effect was a little larger when naming “as hound” than “as dog” (93 vs. 69 ms), although it was significant for both naming tasks, $t > 4.7$, $p < 0.01$.

(b) Synonyms. There were small and mostly non-significant main effects of naming task, $F_1(1, 31) = 0.6$, $F_2(1, 20) = 1.6$, and of relationship, $F_1(1, 31) = 4.3$, $MSe = 2785$, $p < 0.05$, $F_2(1, 20) = 2.97$, $MSe = 1430$, $p < 0.1$; overall, naming times were 12 ms faster in the “as dog” than “as hound” naming task, and 20 ms faster with synonym than unrelated distractors. The important interaction between the two factors was significant, $F_1(1, 31) = 9.1$, $MSe = 870$, $p < 0.001$, $F_2(1, 20) = 6.1$, $MSe = 1113$, $p < 0.05$; the synonym facilitation effect was larger (35 ms) and reliable when naming “as hound”, $t_1(31) = 2.7$, $p < 0.05$, $t_2(20) = 2.8$, $p < 0.05$, and smaller (4 ms) and not significant when naming “as dog”, $t_1(29) = 0.3$, $t_2(20) = 0.3$.

6.1. Discussion

When naming objects with their common (or “dominant”) names, there was facilitation from identical distractor words and no reliable effect of synonym distractors. When performing the novel “naming as hound” task, where rarer alternative synonymous names were produced, there was a large facilitation effect from identical distractors (e.g., HOUND + hound) and a smaller but reliable facilitation effect from the object’s common name presented as a synonym distractor of the target name (e.g., HOUND + dog). The facilitation effects from synonym distractors found in Experiment 4 with monolinguals was remarkably similar in pattern to that observed from translation distractors with bilinguals, if we can accept that the within-language naming “as hound” task is comparable to naming in a less proficient L2 by bilinguals. Bilinguals will have had more general experience separating words in two formally-distinct languages than monolinguals would have particular experience separating words on an *ad hoc* basis in contextually (and here experimentally) determined response sets. But, at the level of each individual lexical selection of a word to say, such separation is often required, for instance to produce a formal word, a familiar alternative form, a specific word, or an informal or slang word (e.g., currency, money, cash, twenty-pound-note, fiver, bread).

Within-language synonyms in monolinguals behaved similarly as cross-language translations did in bilinguals: they exerted no reliable effect when naming in the “more dominant mode” (i.e., naming “as dog” or in L1), but they facilitated naming in the “less dominant mode” (i.e., naming “as hound” or in L2). Our findings regarding the effects of synonym distractors are both novel and have potential theoretical significance, and so before considering their interpretation, it is important to establish their reliability.

7. Experiment 5

The primary purpose of Experiment 5 was to replicate the results of Experiment 4. It used the same design and procedure with two differences. First, the distractors used in the unrelated conditions in Experiment 4 were words matched to the related words on word frequency. Those control words were neither in the “response set” of names the participants were required to produce in an experimental block of trials, nor were they in what might be called the set of “possible object names”. Hermans (2004) argued that the “object set” constitutes a more competitive set of items appropriate for assessing facilitation, and so the distractor words used in the unrelated conditions in

Table 5
Results from English monolinguals in Experiment 4. Mean Object Naming Latencies (in Milliseconds), Standard Deviations, and Rates of Excluded trials (%Ex) by Naming Task and Type of Distractor.

Distractor type	Common naming (as “dog”)			Alternative naming (as “hound”)		
	Mean	SD	%Ex	Mean	SD	%Ex
(a) Identity (congruent)	DOG + <u>dog/eye</u>			HOUND + <u>hound/torch</u>		
Related distractors	807	95	2.2	832	113	2.5
Unrelated distractors	876	96	4.2	925	106	4.6
Facilitation =	69			93		
(b) Synonym (alternative)	DOG + <u>hound/torch</u>			HOUND + <u>dog/eye</u>		
Related distractors	883	99	4.2	881	92	3.9
Unrelated distractors	887	122	3.6	916	119	3.4
Facilitation =	4			35		

Note: Facilitation = Unrelated minus Identity or Synonym.

Experiment 5 were created by re-ordering those in the related conditions (which was also the same method used in Experiments 2 and 3). Second, a slightly different set of stimuli was used.

7.1. Method

7.1.1. Participants

Twenty-four native speakers of English (19 women and 5 men) studying at Essex University volunteered. Most were monolingual, but 7 reported having some proficiency in a second language. Their mean age was 21.1 years (ranging from 19 to 26 years-old).

7.1.2. Stimulus materials

Sixteen pictures were chosen that had high common name agreement but also permitted a close synonym name; 10 of these 16 were also used in Experiment 4 (although GUN was presented with the different synonym *pistol*). The mean rated synonymy of these 16 pairs was 4.35. The common names had higher word frequencies than the synonym names (79 vs. 4 per million). Each picture was paired with: the target name; a word from the set of target names that was re-ordered so as to be unrelated; a synonym of the target name; and a word from the set of synonym names that was re-ordered to be unrelated. The 16 pictures were presented four times in each of the two naming tasks. The same instructions given in Experiment 4 were used, but prior to the “naming as hound” task, each picture was presented three times for familiarization of its target name. Appendix E lists the experimental stimuli used.

7.1.3. Procedure

Participants were tested individually with the same equipment, stimulus appearance, and general procedure as previously. The participants completed two tasks: once naming the pictures with their common names, and once naming them with their lower-frequency synonym alternative names (and the order was counterbalanced). Before each task, the participants were thrice shown the pictures together with their target names, and were asked to produce only these target names. In each task, following the familiarization stage, the participants were given 8 practice trials, followed by 64 experimental trials (16 pictures $\times$ 4 words) presented in a randomized order.

7.2. Results

Responses that were incorrect, hesitant or failed to stop the timer were excluded, as were responses below 250 ms or above 2 s. Overall, 3.9% of responses were excluded, the majority of which were verbal hesitations and stumbles rather than naming errors. Harmonic means of

non-excluded responses were calculated, both for each participant and for each item, in each condition. Table 6 presents the results, which fairly closely replicate those of Experiment 4, both showing that the synonym facilitation effect is found only when naming by producing alternative names (“as hound”). Separate 2×2 related ANOVAs (both by participants and items) were conducted for the identity (or congruency) and the synonym conditions, with the factors of naming task (“as dog” vs. “as hound”) and distractor word relationship (related vs. unrelated).

(a) *Identity*. There were main effects of naming task, $F_1(1, 23) = 20.3$, $MSe = 8079$, $p < 0.0001$, $F_2(1, 15) = 41.3$, $MSe = 3199$, $p < 0.0001$, and distractor relationship, $F_1(1, 23) = 67.7$, $MSe = 3534$, $p < 0.0001$, $F_2(1, 15) = 98.3$, $MSe = 1579$, $p < 0.0001$. Participants were 83 ms faster to name pictures with their common than their alternative synonym names, and were 100 ms faster with identical than unrelated distractors. The interaction between the two factors was marginal, $F_1(1, 23) = 3.9$, $MSe = 1272$, $p < 0.1$, $F_2(1, 15) = 2.3$, $MSe = 1793$, but the trend was for the identity facilitation to be larger for the naming “as hound” task (although it was significant for both, $t > 5.0$, $p < 0.01$).

(b) *Synonyms*. There were significant main effects of naming task, $F_1(1, 23) = 34.4$, $MSe = 3994$, $p < 0.0001$, $F_2(1, 15) = 45.0$, $MSe = 2099$, $p < 0.0001$, and distractor relationship, $F_1(1, 23) = 21.8$, $MSe = 1070$, $p < 0.0001$, $F_2(1, 15) = 16.6$, $MSe = 927$, $p < 0.001$; naming times were 76 ms faster in the “as dog” than the “as hound” naming task, and were 31 ms faster naming with synonym than unrelated distractors. Importantly, the interaction between the two factors was also significant, $F_1(1, 23) = 6.0$, $MSe = 1589$, $p < 0.05$, $F_2(1, 15) = 8.4$, $MSe = 968$, $p < 0.01$. The synonym facilitation effect was significant when naming “as hound”, $t_1(23) = 5.4$, $p < 0.01$, $t_2(15) = 5.0$, $p < 0.01$, but not when naming “as dog”, $t_1(29) = 1.2$, $t_2(15) = 0.8$.

7.3. Discussion

Experiment 5 replicated the results of Experiment 4. Synonym distractors produced facilitation when producing alternative names (in our novel “name as hound” task), but had no reliable effect when producing common names (in the standard “name as dog” task). We explain the absence of synonym facilitation when producing common names in a manner similar to our account of the smaller translation facilitation when bilinguals name in L2. We propose that the lexical representations of close synonyms that function as alternative names are interconnected in the speech production system, but that the alternative-to-common name connections are weaker than vice versa; see General Discussion.

In both Experiments 4 and 5, naming times were slower when producing alternative names than common names. This is not particularly surprising. Word frequency may certainly play some role in this difference, as the alternative names were generally rarer than the common names (e.g., “automobile” vs. “car”). It is unlikely that word frequency alone is the sole explanation however. The word frequencies of category names such as “animal” and “vehicle” are generally higher than those of names of objects within the categories, and yet producing category-level names is slower than basic-level naming. In the picture-word task, Costa, Mahon, Savova, and Caramazza (2003) found that the same pictures (with the same distractor words) were overall 64 ms slower to name with category-level than basic-level names (e.g., “animal” and “tool” vs. “cat” and “hammer”). Vitkovitch and Tyrrell (1999) found that participants were overall 119 ms slower to name the same set of pictures with subordinate-level than basic-level names (e.g., “poodle” and “adder” vs. “dog” and “snake”). The synonymous alternative names used in Experiments 4 and 5 are less “dominant” than the common names. Indeed, we selected objects whose alternative names were almost never produced in a free naming task. However, perhaps similar to a low-frequency L2 word, a familiar synonym is an additional, alternative name for the same thing.

Table 6

Results from English monolinguals in Experiment 5. Mean Object Naming Latencies (in Milliseconds), Standard Deviations, and Rates of Excluded trials (%Ex) by Naming Task and Type of Distractor.

Distractor type	Common naming (as “dog”)			Alternative naming (as “hound”)		
	Mean	SD	%Ex	Mean	SD	%Ex
(a) Identity (congruent)	DOG + <i>dog/barrel</i>			HOUND + <i>hound/cask</i>		
Identity	760	117	2.8	828	97	2.5
Unrelated	845	126	4.2	942	101	4.8
Facilitation =	85			114		
(b) Synonym (alternative)	DOG + <i>hound/cask</i>			HOUND + <i>dog/barrel</i>		
Synonym	830	95	2.3	887	99	3.8
Unrelated	842	104	5.4	938	107	5.0
Facilitation =	12			51		

Note: Facilitation = Unrelated minus Identity or Synonym.

In our novel alternative synonymous name production (or “naming as hound”) task, participants were instructed to use names of the stimulus objects that they would not produce spontaneously. The latencies to produce these non-preferred names were faster when the preferred names were presented as synonym distractors (e.g., SWINE + pig). Hantsch and colleagues (e.g., Hantsch, Jescheniak, & Mädebach, 2012; Hantsch, Jescheniak, & Schriefers, 2005; Hantsch & Mädebach, 2013) report experiments using a superficially similar task that produced quite different results. They presented pictures whose preferred names were at a sub-ordinate level (e.g., “shark”, “rose”, and “Mini”), but required participants to name them with basic-level names (e.g., as “fish”, “flower”, and “car”). When the preferred names were presented as auditory distractors (e.g., FISH + “shark”), these produced interference. Clearly, producing non-preferred names cannot by itself be responsible for the polarity of the effect of synonym distractors we observed. We submit that our synonyms are neither superordinate names nor necessarily the names of more precisely specified different subordinates, but are same-level alternative names of the same thing.

Also of relevance to our findings are two other interesting studies using objects with near-synonymous names that suggest that the phonological forms of both names are co-activated. Peterson and Savoy (1998) investigated the phonological encoding of names of pictures that had both a primary and a secondary name, such as “couch” or “sofa” and “pig” or “hog”. Participants were required to name pictures, but on trials when a printed word was presented superimposed upon a picture, they were required switch task to name the word. The target words were phonologically related to the primary name of the picture (e.g., count, peg), phonologically related to the secondary name (e.g., soda, hug), or unrelated to either, and were presented at various stimulus-onset asynchronies (SOAs). Word naming times were faster when the words were phonologically related to either the primary or the secondary names of the object, and this effect was observed for SOAs of 150, 200, 300 and 400 ms. At an SOA of 50 ms, there were no effects at all, and at an SOA of 600 ms only words phonologically related to the primary names showed facilitation. These results suggest that the phonological representations of both an object’s alternative (or “near-synonymous”) names are activated and so prime word reading-times. Levelt et al. (1999) acknowledge that simultaneously selecting multiple lexical representations can occur under time pressure. Our results show that, at the lexical level, alternative names are routinely co-activated.

In a picture-word task, Jescheniak and Schriefers (1998) presented auditory distractor words that were phonologically related either to the target names (e.g., BURG (castle) + “Bursche”) or to near-synonym names (e.g., BURG + “Fenchel”, which is similar to “Festung”, an alternative name of a castle). Naming times were facilitated by distractors phonologically related to the target, but were slowed by distractors phonologically related to the alternative synonymous name. This interference effect has clear parallels with the “phono-translation” interference effects in bilinguals (Costa et al., 2003; Hermans et al., 1998), where distractors phonologically related to non-target translations slowed target naming. The combination of these results supports the notion that both monolingual and bilingual speakers activate both (or all) of a stimulus object’s alternative names. If we accept the claim that a distractor’s phonological overlap with an alternative name will make it a stronger competitor (that will consequently slow target naming), it is therefore surprising that when the alternative name itself is presented as a distractor, naming was facilitated and not impeded. Surely the whole word *hound* should make ‘hound’ a much stronger competitor to the target ‘dog’ than a phonologically related word such as *how*. We found that the alternative names themselves presented as distractors facilitated naming times, and submit that these results provide more direct evidence concerning lexical selection than the indirect, phonologically-mediated effects that may operate post-lexically.

7.3.1. The nature of synonyms

For various historical and etymological reasons, a language may

contain different words that mean the same thing, although the existence of synonyms may seem pointless, futile, and unnecessary. The words we chose as synonyms of object names were considered to be plausible alternative names of our stimulus line drawings. Further, independent participants gave these words high ratings for their degree of synonymy (or “closeness of meaning”) to the object names, with means of 4.5/5 in Experiment 4 and 4.4/5 in Experiment 5. But were they “true” synonyms? Some were reasonably interchangeable (e.g., glasses–spectacles; sofa–couch) or had closely overlapping, very similar referents (e.g., ghost–phantom). However, it is undeniable that some were “less identical”: some were like specific subordinates (e.g., ammunition–bullet, dog–hound), and some were equivalent only in terms of their relationship to our specific pictured stimuli (e.g., clock–time-piece, crocodile–alligator). More generally, Clark (1987) argued that, because “any difference in form in a language marks a difference in meaning” then “there are no true synonyms” (p. 3). Social-linguistic processes operate to differentiate the meanings and connotations of synonyms by using one word in different contexts or with slightly different referents (e.g., an automobile coming to be used to refer to a particular type of car). It is certainly true that our synonyms were not semantically interchangeable in all communicative contexts. However, we must emphasize that the synonyms we used were entirely acceptable alternative names of the pictured objects presented as stimuli.

7.3.2. Effects of the order of naming with common and alternative names

In our two experiments with monolinguals, half of the participants first named the objects with common names and then with alternative names, whereas half had the opposite order. To examine the effect of order of naming task, we combined the data from the two experiments and performed similar analyses to those performed on the bilingual data. The results are shown in Table 7, and were analysed by two mixed 3-way ANOVAs (by participants only), separately for identity and synonym distractors, with the between-participant factor of order of naming task (common first vs. alternative first) and the within-participant factors of naming task and distractor relationship.

(a) Identity. The analysis repeated effects previously reported for each experiment: there were significant main effects of naming task, $F_1(1, 54) = 24.3, p < 0.001$, and distractor type, $F_1(1, 54) = 125.2, p < 0.0001$, and a significant interaction between naming task and distractor type, $F_1(1, 54) = 7.4, p < 0.01$; the facilitation effect from identical words was larger for alternative than for common naming (101 vs. 76 ms).

There was a significant main effect of order of naming task, $F_1(1, 54) = 4.5, p < 0.01$; participants who produced common names first were 46 ms faster than those who produced alternative names first. The significant interaction between order and naming task, $F_1(1, 54)$

Table 7
Effects of Order of Naming Task in Monolinguals. Mean Object Naming Latencies (in Milliseconds) collapsed over the participants tested in Experiments 4 and 5, by Order of Name Production, Naming Task, and Type of Distractor.

Naming task =	Common naming first		Alternative naming first	
	Common	Alternative	Common	Alternative
<i>(a) Identity (congruent)</i>				
Identity	729	842	844	819
Unrelated	808	940	917	924
Facilitation =	79	98	73	105
<i>(b) Synonym (alternative)</i>				
Synonym	832	883	889	884
Unrelated	820	927	915	924
Facilitation =	– 12	44	26	40
Mean naming time	797	898	891	888

Note: Facilitation = Unrelated minus Identity or Synonym. Mean naming time = mean of each naming task over all four distractor types.

= 32.9, $p < 0.001$, showed that participants who produced common names first were 122 ms faster producing common than alternative names, whereas those who produced alternative names first showed only a small difference (of 9 ms) between producing common and alternative names. The interaction between order and distractor type was absent, $F < 1$.

(b) *Synonym*. The analysis also repeated effects previously reported for each experiment. There were significant main effects of naming task, $F_1(1, 54) = 12.9$, $p < 0.001$, and distractor type, $F_1(1, 54) = 16.6$, $p < 0.001$, and a significant interaction between naming task and distractor type, $F_1(1, 54) = 16.0$, $p < 0.001$; the synonym facilitation effect was larger for producing alternative than common names (42 vs. 7 ms).

Participants who produced common names first were 37 ms faster than those who produced alternative names first, but this main effect of order of naming task did not reach significance, $F_1(1, 54) = 2.4$. The significant interaction between order and naming task, $F_1(1, 54) = 11.9$, $p < 0.01$, showed that participants who produced common names first were 79 ms faster producing common than alternative names, whereas those who produced alternative names first were only 2 ms faster producing common than alternative names. The interaction between order and distractor type was not significant, $F_1(1, 54) = 1.9$; the synonym facilitation effect was only slightly smaller for participants who produced common names first (16 vs. 33 ms).

Discussion. These analyses show that the facilitation effects from identical and synonym distractor words were not affected significantly by the order of presentation of the two naming tasks. This attests to the generality of the effects we have established in the picture-word task. Although order of presentation of the two naming tasks had no effect at all on the synonym facilitation effect when producing alternative names, there was a non-significant ($t = 1.03$) tendency for synonyms to produce a small interference effect (of 12 ms) when producing common names. This aspect of the results will require replication and further study before it can support any theoretical interpretation.

The overall naming times of monolingual speakers showed asymmetric after-effects from having previously produced different names for the same objects. Participants who first produced common names (such as “dog”) were faster producing common than alternative names, whereas participants who first produced alternative names (such as “hound”) showed no reliable difference between producing common and alternative names. Interestingly, these asymmetric naming after-effects with monolingual speakers parallel the findings of Misra et al. (2012) and Branzi et al. (2014) with bilinguals (although not the results of our experiments with bilinguals).

8. General discussion

Both cross-language translations and within-language synonyms are alternative names for the same object or concept. As the lexical representations of translations and synonyms are extremely closely related in meaning, both face the same critical “hard problem” of lexical selection (Finkbeiner et al., 2006). The purpose of our work was to investigate the extent of similarities of the effects of translation distractors for bilinguals and of synonym distractors for monolinguals in the picture-word task. Our experiments found strikingly similar patterns of results, in two major respects. First, robust identity facilitation effects were found consistently, both when bilinguals name in L1 and in L2, and when monolinguals produce common names and alternative synonymous names of objects. Second, there was facilitation—and never interference—from distractor words that were translations or synonyms of the target. The translation facilitation effect showed a language asymmetry: it was larger when bilinguals named in L2 than in L1. The synonym facilitation effect showed a similar asymmetry: it was larger (and only significant) when monolinguals produced alternative

rather than common names. The fact that translations in bilinguals and synonyms in monolinguals produce parallel patterns of results is, we contend, impressive and strongly motivates seeking a theoretical framework that encompasses both the commonalities of monolingual and bilingual lexical selection as well as the particular constraints of bilingualism.

One possible account of the similarity of results is that both translation and synonym facilitation effects should be understood in terms of word frequency effects. Many studies of object naming have found that overall naming times in bilinguals are slower, both in L1 and L2, than in monolinguals (e.g., Francis, Corral, Jones, & Sáenz, 2008; Gollan, Montoya, Cera, & Sandoval, 2008). This is easily explained by the “frequency lag hypothesis” of Gollan et al. (2011). Assuming that bilinguals produce the same total number of words as monolinguals, then the fact that they use two languages to produce words means that the frequency of use of any particular word in a language will be functionally reduced. We did not find consistently reliable main effects of naming language in our experiments with bilinguals, which may have been due to our methodology, as each object was both familiarized and repeated four times in each naming language. In our experiments, the related (i.e., identical, translation and synonym) distractor words were carefully matched to unrelated control words on word frequency, and so the frequency of distractors cannot explain the patterns of results we report in the picture-word task. The primary purpose of our work was to compare the pattern of effects of the experimental manipulation of type of distractor words within groups of bilingual and monolingual speakers. We maintain that the pattern of results we have demonstrated within (and replicated between) groups of participants are unlikely to be due solely to effects of frequency, although future research will be required to establish this with confidence.

8.1. Identity vs. translation and synonym facilitation

Substantial facilitation was observed from distractor words that were identical to target names. This congruency effect is not particularly surprising; indeed, it may even be inappropriate to call identical words “distractors” at all. The identity facilitation effect could result from priming at any or all levels of target name activation, selection, and retrieval (i.e., at semantic, lexical, and phonological processes), as well as eliminating any possible “conflict surprise” from the simultaneous presentation of different words and pictures. The facilitation from identical distractors was always larger than from either translations in bilinguals or synonyms within a language; indeed, translation and synonym facilitation was about one-quarter to a-half the magnitude of identity facilitation. Costa et al. (1999) argued that the identity effect reflected contributions from both semantic and sublexical phonological processes, whereas the translation effect resulted only from word-to-target semantic priming. If semantic-level processing were to be solely responsible for the translation effect, then it is not clear why there should be language asymmetry in the facilitation effect we observed. The translation facilitation effect was larger when naming in L2 (e.g., DOG + perro) than in L1 (e.g., PERRO + dog), but there is no evidence to suggest that perro would prime “dog” more than dog would prime “perro”. It is possible that L2 printed words do not access their semantics as rapidly as L1 words, and/or that the lexical phonological representations of L2 words are less strong or less well established than those of L1 words. However, without more direct evidence to assess these possibilities, our current preference is to propose a lexical-level rather than a solely semantic-level account of the language asymmetry of the translation facilitation effect.

The new theoretical claim we advance, the inter-lexical translation connection hypothesis, concerns the organization of the lexical representations of translations. We propose that the lexical representations of translations are inter-connected by facilitatory links, although

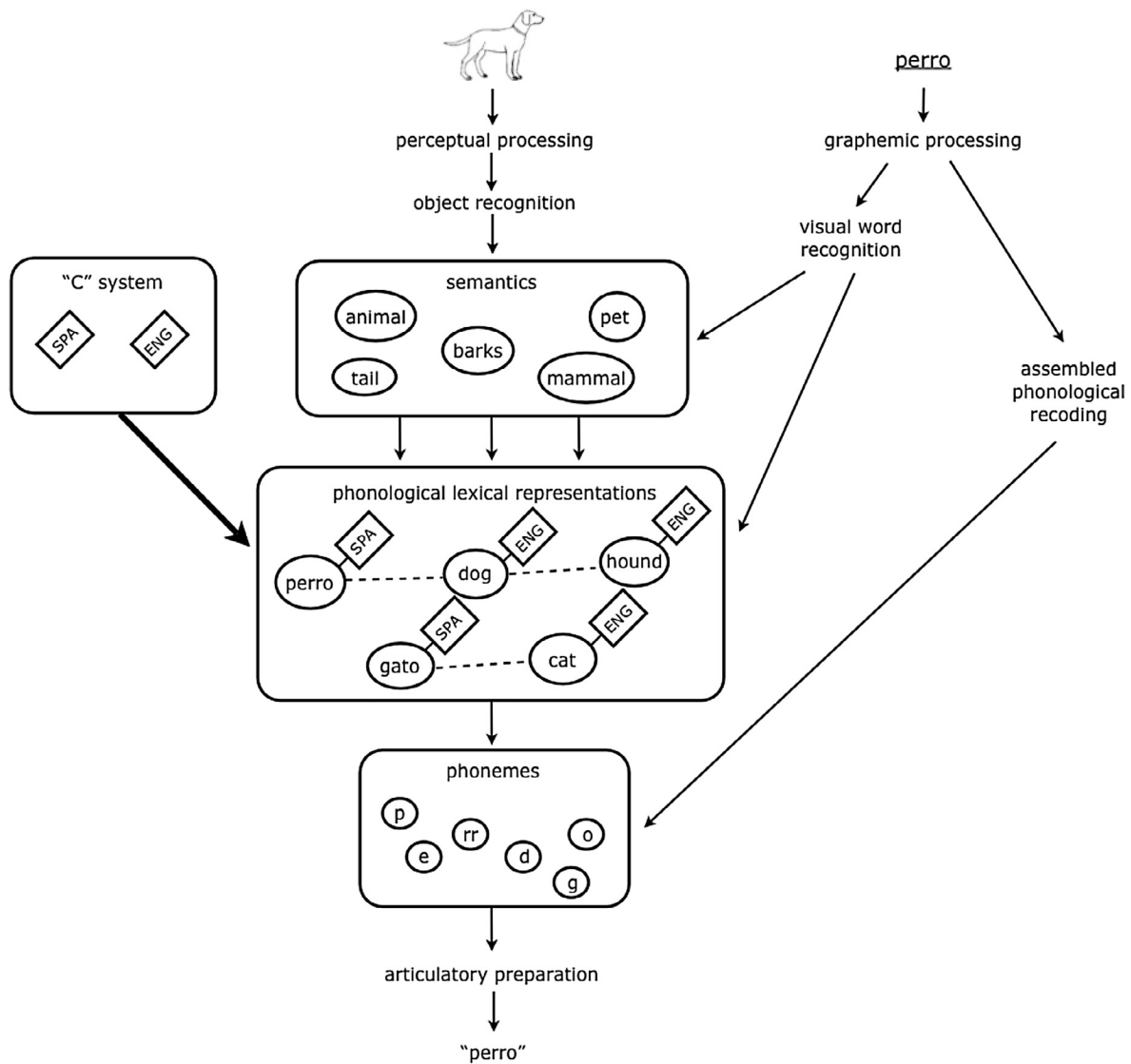


Fig. 1. A framework for understanding the processes involved in the picture-word task for bilinguals and monolinguals. Lexical representations are shown as having “language tags” which can be activated by the supervisory C-system.

the lexically specific L1→L2 connections are assumed to provide more facilitation than L2→L1 connections. For example, for a bilingual with L1 Spanish and L2 English, the ‘perro’→‘dog’ connection will be stronger than the ‘dog’→‘perro’ connection. Before incorporating this hypothesis within a broader theoretical framework for understanding bilingual lexical selection, we shall first offer a specified framework of the points of contact where the reading system may impact upon the processing sequence responsible for object naming, and then consider the implications of our results for theories of the role of competition in monolingual lexical selection.

Fig. 1 presents a theoretical framework of the processing streams underlying object naming (e.g., Humphreys, Riddoch, & Quinlan, 1988; Morton, 1985) and word reading as postulated by the “three-route” model (e.g., Coltheart, Rastle, Perry, Langdon, & Ziegler, 2001; Morton, 1979). Importantly, Fig. 1 makes explicit at what points these two processing streams come into contact (which has often been lacking in accounts of results from the picture-word task). Levelt (1989) and Levelt et al. (1999) proposed a distinction between an abstract lemma

and a phonological lexeme representation of words. The existence of a separate lemma level has been criticized (e.g., Caramazza, 1997) and Fig. 1 avoids the “lemma dilemma” debate (Rapp & Goldrick, 2006) by showing only phonological lexical representations.

The semantic reading route applied to printed words *perro*, *hund*, *dog*, and *chien* will activate their common semantic representation. Semantic activation is passed to the lexical representations of both L1 and L2 names, and so both should be equivalently semantically primed by a translation distractor word. The direct lexical reading route activates the phonological lexical representation of the specific word presented (e.g., *perro* will activate ‘perro’ but not ‘dog’ directly). For identical distractors, the direct lexical reading route will activate the target name, which contributes to the identity facilitation effect. For translation distractors (e.g., DOG+*perro*), the direct lexical reading route will activate the non-target translation. (Rather than increasing its status as a competitor, which should lead to interference, we found that translations produce facilitation.) Finally, the phonological recoding reading route activates sub-lexical phoneme units. If there exists

feedback from the phoneme-level to the lexical-level, as some theories suggest (e.g., Goldrick & Rapp, 2002), then sub-lexical phonology may also affect lexical selection indirectly.

8.2. Lexical selection by competition?

The standard explanation of interference effects in the Stroop and picture-word interference tasks is that response selection involves competition between activated related words. Accounts of results from the picture-word task have been offered within general theories of lexical competition in speech production (e.g., Levelt et al., 1999). A stimulus object will activate a number of related words. If a distractor additionally activates one of these non-target words, then it will become a closer competitor and so slow target name selection. In contrast, Mahon et al. (2007) proposed the response exclusion hypothesis as a non-competitive account of performance in Stroop-like tasks. The response exclusion hypothesis (Finkbeiner & Caramazza, 2006; Janssen, Schirm, Mahon, & Caramazza, 2008; Mahon et al., 2007) proposes that performance in the picture-word task reflects the operation of a post-lexical and non-competitive mechanism that excludes “production ready” representations of non-target words from a single-channel speech output buffer.² It is assumed that presented words have “privileged” access to the articulators, and so a printed distractor word will generally enter the output buffer before the target object’s name. The buffer must therefore be cleared (or purged) of the non-target word before the target name can be articulated. A number of “response relevant criteria” operate to exclude non-target words from the buffer. Mahon et al. propose that “response level representations can index general properties of their corresponding concepts (e.g., semantic category) as well as their source (picture or word)” (p. 524). Distractor words that match (or satisfy) response relevant criteria are slower to exclude from the output buffer, which delays target naming time. Category coordinate distractors (e.g., BED + chair) will match a “same category as target object” criterion and so will be slower to exclude from the buffer, which produces the often replicated semantic interference effect. Semantically related words from a different category (e.g., LION + Africa) will not match the “same category as target object” criterion and so will be excluded faster; this faster exclusion then allows distractor-to-target semantic priming to be detected as a facilitation effect (e.g., de Zubicaray et al., 2013). All distractor words will be ultimately excluded by a “source” criterion: for a response to pass through the buffer to produce in speech, it must satisfy the “name the object, not the distractor” task-required criterion. The fact that semantic interference is observed at all in the picture-word task suggests that this “source” criterion cannot be the first to be applied.

There are two findings of Experiments 4 and 5 that are directly relevant to the theoretical debate between the lexical selection by competition theory and the response exclusion hypothesis: (i) In the basic naming task of producing common names, there was no interference from synonym distractors (e.g., DOG + hound), and (ii) in the task of producing alternative names to the same objects, there was reliable facilitation from synonym distractors (e.g., HOUND + dog). Both findings are problematic for the selection by competition theory (and are certainly contrary to the spirit of its approach) as the theory predicts that synonym distractors, as very closely related words, should interfere with target selection. For example, for DOG + hound the non-target lexical representation ‘hound’ will be activated (both by the semantic system and by the direct lexical reading route) almost as highly as the target representation ‘dog’, making it a very close competitor that should impede target selection. The reliable synonym facilitation effect

in the “naming as hound” task is even more difficult for the selection by competition theory to explain than the absence of an interference effect. For example, for HOUND + dog the selection by competition theory must surely expect that the non-target lexical representation of the more dominant name ‘dog’, activated by both the semantic system and the reading routes, will be an extremely close competitor that should interfere with target selection.

When the word dog is presented as a synonym distractor in the “naming as hound” task (i.e., when participants are instructed *not* to produce common dominant names), it would not be a member of a task-determined “response set”. (The common names would remain part of the set of “possible names” of the stimulus objects, in much the same way that translations for bilinguals can be seen as alternative names; see our earlier discussion of Hermans, 2004.) Effects of response set may be explained by proposing that the activation levels of target names are selectively enhanced; for example, when naming “as hound”, the activation of ‘hound’ would be boosted relative to that of ‘dog’ (which would continue to be a lexical competitor). However, this account predicts that HOUND + dog should produce interference whereas our experiments found a facilitation effect.

A possible option for a modified selection by competition account of our observed synonym facilitation effect would be to propose that words not in the response set might be inhibited or somehow suppressed by the task instruction to name only “as hound” (in order to prevent the production of dominant common names). Such suppression would have to operate solely at the level of output phonology, as the printed word dog facilitates naming the object as “hound”, by priming at the semantic level. It is not at all obvious how such a modified account would be able to execute suppression of words not in the response-set without sacrificing the basic notion of selection by competition operating on each individual naming trial.

The response exclusion hypothesis requires extra and post hoc specification to explain the results of our experiments. In the case of synonym distractors in common naming (e.g., DOG + hound), the exclusion of ‘hound’ from the output buffer should be slowed by it matching the “same category as target object” criterion, although it would not match a task-determined criterion of “in the response set” (which would be contextually specific to the naming task undertaken in the experiment). It might be possible for the response exclusion hypothesis to argue that the effects of matched and unmet criteria cancel each other out, but it will need to make explicit and empirically supported proposals concerning how response relevant criteria are applied (e.g., in a serial order determined by their level of generality?) and what are their relative weightings.

In the case of synonym distractors in the naming “as hound” task, the response exclusion hypothesis must claim that a new and important response relevant criterion comes to be established quite prominently: there is the salient task requirement to not produce the common names of the stimulus objects. Although the exclusion of the distractor word ‘dog’ would be slowed by it matching the “same category as target object” criterion, it should be excluded faster by not matching the task-determined “in response set” criterion. Assuming that this “response set” criterion operates more strongly (or is applied earlier) than the “same category” criterion, then the speedier exclusion by a word violating the “response set” criterion should allow the strong semantic-level priming from the distractor word to the processing of the target to manifest itself in the overall speeding of target naming. This line of reasoning ascribes quite a powerful role to the task-specific “response set” criterion, and re-raises the under-specification of the response exclusion hypothesis in terms of how response relevant criteria are applied (e.g., sequentially or in parallel), and whether there are differential effects of multiple criteria on overall naming times. The account constructed here for how the response exclusion hypothesis might interpret the HOUND + dog facilitation effect relies on the notion that the general “in the set of target synonym name responses” task criterion must be more influential than the “same category as target” criterion

² This hypothesis does not appeal to competition in the sense of a “battle” that harms or slows all activated combatants, but it does entail competition in the sense of a “race” to the speech output buffer where many winners (i.e., distractor words) must be disqualified.

that must be set for each individual stimulus object on a trial-by-trial basis.

In summary, the results of our experiments provide no clear support for theories of lexical selection that propose competition between the representations of co-activated related words, as these must predict semantic interference from closely related synonyms. The selection by competition theory takes semantic interference effects as indisputable evidence for lexical competition. The response exclusion hypothesis is geared-up to more readily account for facilitation than interference effects, but it will require more detailed specification to provide a satisfactory account of our results. Hall (2011) discussed how the response exclusion hypothesis may be applied to bilingual performance in the picture-word task (see also Finkbeiner et al., 2006), but concluded that it cannot account for the conjunction of findings that cross-language semantically related distractors produce interference (e.g., PERRO + wolf; Costa et al., 1999; Smith & Kirsner, 1982), whereas translation distractors (e.g., PERRO + dog) produce facilitation.

8.3. Bilingual lexical organization and selection

The framework depicted in Fig. 1 presents bilingual lexical representations as having “language tags” (Green, 1998; Poulisse, 1997) rather than existing within separate lexicons for each language (e.g., Costa et al., 1999); however, both formulations are formally identical means of allowing language-specific lexical selection. Fig. 1 also depicts our inter-lexical translation connection hypothesis of the organization of the lexical representations of translations, by showing that specific connections exist between the lexical representations of translation equivalents (e.g., ‘perro’ ↔ ‘dog’). More speculatively, we have also shown specific connections between the lexical representations of close synonyms in monolinguals (e.g., ‘dog’ ↔ ‘hound’).

Fig. 1 also presents an extra-lexical executive control mechanism, which we call the “C system” to indicate language control, that can activate lexical representations via their associated language tags. Such a system is necessary to enable bilinguals to select words to produce in one language only, either by intention within spontaneous speech or by compliance with specific task requirements. Our conception of this system is similar to the purpose of Green’s (1998) “language task schemas”, Poulisse and Bongaerts’ (1994) “conceptual cues”, and de Bot’s (2004) “external language nodes”. (So, “C” stands for control, choice, conceptual cue, and communicative context.) In addition to enabling bilinguals to choose to speak in one language, the C-system will also enable them, when required or desired, to purposefully translate words and to switch between languages. All of these skills are essential aspects of language competence in bilingualism. However, we also conceive of the C-system in more general terms. It is the general executive mechanism that enables all speakers to implement a range of behavioural and social intentions, such as complying with a particular task (e.g., to name the object and not the printed word in the picture-word task) and selecting contextually appropriate “registers” of speech (e.g., when to use formal vs. informal words). As such, the C-system may supervise some of the functions that Mahon et al. (2007) ascribe to response relevant criteria and some of La Heij’s (2005) proposals concerning how message-level communicative intent is executed lexically.

Our interpretations of our picture-word results within the framework presented in Fig. 1 rely upon four key assumptions. First, the C-system increments the activation levels of lexical representations in the target language and satisfy the task requirement of naming the object and not the distractor word. As the C-system provides two strong sources of activation to target names, it will provide more activation than from either the semantic system or the reading routes (and hence the thicker lines from it in Fig. 1). Second, the semantic system will

activate the lexical representations of both target words and their translation equivalents. (Differences in the relative balance of L1 and L2 activation levels are likely to vary due to the degree of L2 proficiency and/or language immersion.) Third, the reading system will activate (i) the semantics of the distractor word (by the semantically-mediated route), which then passes activation to both L1 and L2 lexical representations; (ii) the lexical representation of the distractor word itself (by the direct lexical reading route); and (iii) the phoneme-level representation of the distractor word (by the assembled phonological recoding route). The direct (but non-semantic) lexical reading route, which must be language-specific for non-cognate words, is crucial for our reasoning; for example, the distractor word *dog* will activate the lexical representation ‘dog’ but have no direct impact on its translation ‘perro’.

Finally, the new proposal in this account, our inter-lexical translation connection hypothesis, proposes that there are specific connections between the lexical representations of translation equivalents. We conceive of these connections as being essentially bi-directional, but, for speech production, there will be stronger L1-to-L2 than L2-to-L1 connections such that the “net flow” of activation will be greater from a L1 representation to its corresponding L2 translation. We therefore expect that the net flow will be more apparent when the task is to name in L2 (where the translation facilitation effects are larger). We appreciate that this proposed net flow of activation is in the opposite direction to that suggested by Kroll and Stewart’s (1994) “revised hierarchical” model. We accept that, for word comprehension, bilinguals are likely to benefit from stronger L2-to-L1 than L1-to-L2 connections in the visual word recognition system, as these will allow faster translation into their more dominant L1 that is more strongly connected to the semantic system. However, we maintain that, for naming, bilinguals benefit from stronger facilitation from their L1 to their less dominant L2 lexical representations responsible for speech production.

To interpret the bilingual results of Experiments 1, 2 and 3, Fig. 2a–d shows particular instances of the broad framework to illustrate the sources of activation of the lexical representations of the translation-equivalents ‘perro’ (L1) and ‘dog’ (L2) for Spanish-English bilinguals. In particular, these illustrate the extra activation provided by the reading routes from related distractor words and from our inter-lexical translation connection hypothesis. We assume that the selection of the target’s lexical representation will be accelerated (and so facilitated) if it is differentially more activated than that of unrelated control words.

Fig. 2a depicts the situation when naming is in L1 and the distractor is the same name as the target (e.g., PERRO + perro). The activation of the target lexical representation ‘perro’ will be increased by the C-system and by the direct lexical reading route (and the phonemes of “perro” will also be activated by the assembled phonological recoding route). The sum of this increased activation results in the facilitated selection and production of the target “perro”. A very similar situation occurs when naming is in L2 and the distractor is identical to the target name (e.g., DOG + dog), as depicted in Fig. 2c, where there will also be some activation of the target from the lexical connection between ‘perro’ ↔ ‘dog’.

Fig. 2b depicts the case when naming is in L1 and the distractor is the target’s L2 translation (e.g., PERRO + dog). The C-system will increase the activation level of the target lexical representation ‘perro’, and the direct lexical reading route will increase the activation of ‘dog’ (and the assembled phonological route will activate the phonemes of “dog”). For bilinguals with high levels of L2 proficiency, our inter-lexical translation connection hypothesis proposes that some activation will be passed along the weaker ‘dog’ → ‘perro’ connection (depicted as a dotted line in Fig. 2b) that results in an overall facilitation for naming

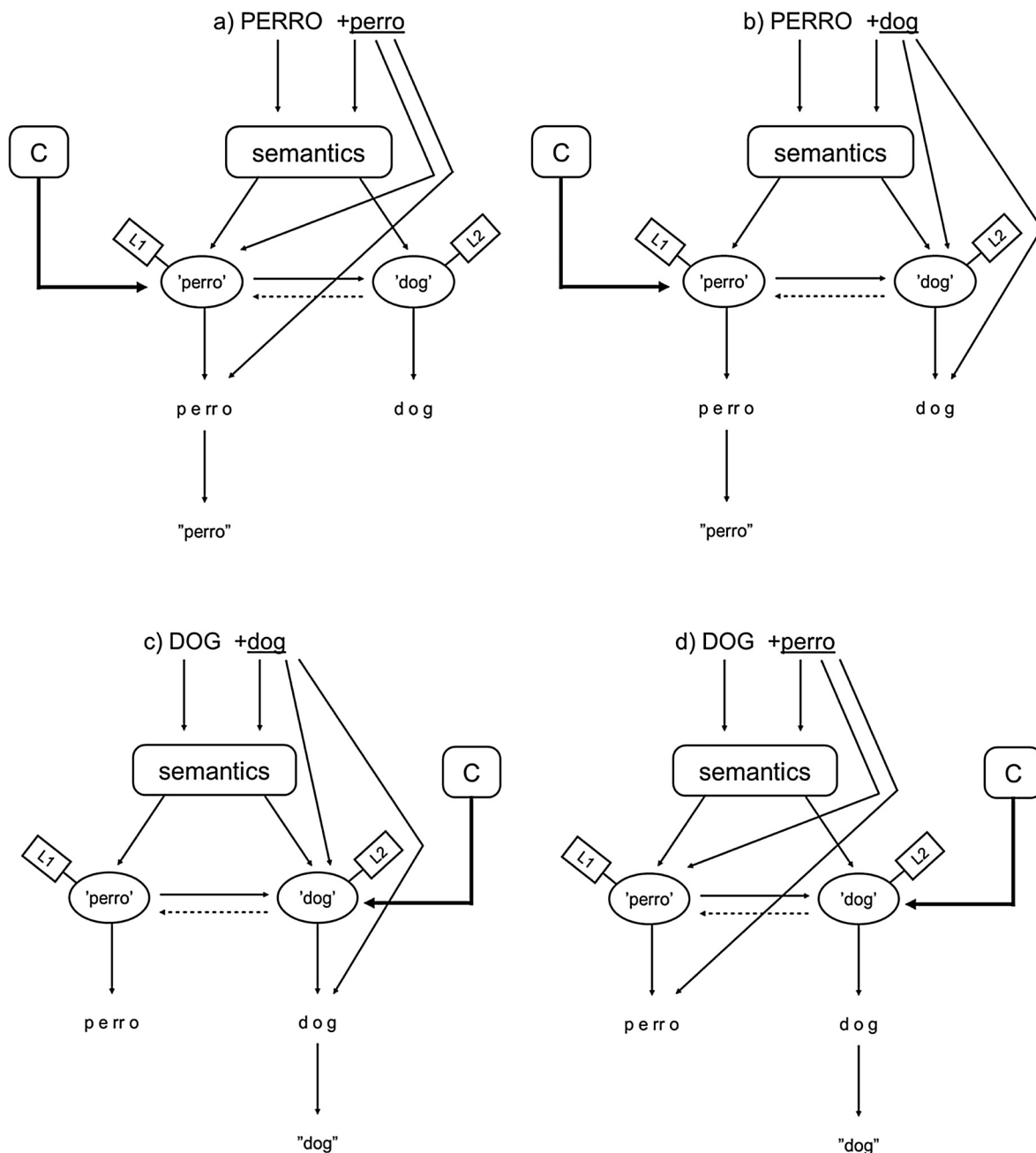


Fig. 2. The use of the framework in Fig. 1 to interpret the facilitation effects observed in the present studies, using the example of *perro* (L1) and *dog* (L2) for Spanish-English bilinguals. The C-system activates lexical representations in the target language that also satisfy task requirements. As it provides two sources of activation, this figure shows thicker lines from it than from the semantic system or the reading routes. The L1-to-L2 ('perro'→'dog') connection (shown by a solid line) is proposed to be stronger than the L2-to-L1 ('dog'→'perro') connection (shown by a dotted line).

"perro".

Fig. 2d depicts the case when naming is in L2 and the distractor is the target's L1 translation (e.g., DOG + perro). The C-system will increase the activation of the target 'dog', and the direct lexical reading route will increase the activation of 'perro' (and the assembled phonological route will activate the phonemes of "perro"). Our inter-lexical translation connection hypothesis proposes that the 'perro'→'dog' connection will also increase the activation of the target 'dog', which will enjoy an activation advantage over unrelated words and so

facilitate its selection. As we propose that there are stronger L1-to-L2 ('perro'→'dog') than L2-to-L1 ('dog'→'perro') connections, the facilitation effect for DOG + perro will be greater than that for PERRO + dog (see Fig. 2b); this explains our observed asymmetry in the translation facilitation effect. The magnitude of the facilitation in DOG + perro will be less than when the distractor word is identical to the target name (in DOG + dog, see Fig. 2c), where 'dog' will be strongly activated by both the C-system and the direct lexical reading route.

8.4. The lexical organization of alternative names

In order to interpret the results of Experiments 4 and 5, we need only to substitute “common name” for L1 and “alternative synonymous name” for L2 into the accounts we have just presented. We speculate that there are specific connections between the lexical representations of close synonyms (i.e., alternative object names) in the speech production system, with a greater net flow of activation from common names to their rarer synonym alternatives (e.g., ‘dog’→‘hound’). This proposal parallels our inter-lexical translation connection hypothesis advanced to account for our results from bilinguals.

Both Experiments 4 and 5 found that when producing common names of objects there was no reliable facilitation from synonym distractors (e.g., DOG + hound). Our explanation of this result parallels our accounts of when bilinguals name in L1 with L2 translation distractors (see Fig. 2b). The C-system will increase the activation of the target lexical representation ‘dog’ (as the task is to produce common names), and the direct lexical reading route will increase the activation of the distractor ‘hound’. However, the target ‘dog’ will not benefit from the alternative-to-common name (‘hound’→‘dog’) connections that are assumed to be considerably weaker than the common-to-alternative (‘dog’→‘hound’) connections. (Our interpretation of the reliable synonym facilitation effect when producing alternative names parallels our account of when bilinguals name in L2 with L1 translation distractors, see Fig. 2d, where the stronger ‘dog’→‘hound’ connections come more to the fore.) In our interpretation of the translation facilitation of naming in L1 (e.g., CHIEN + dog) found in Experiments 2 and 3, we argued that as L2 lexical representations strengthen with increasing proficiency, they will produce reverberating activation between translations, that will serve to facilitate naming in both L1 and L2. Similarly, we predict that synonym distractors would facilitate naming with common names as the relative strengths of the bi-directional connection between synonyms become more “balanced”. This could be tested experimentally. The prior and repeated reading aloud of lower-frequency synonyms (e.g., spectacles) should produce a facilitation effect when these are subsequently presented as distractor words in the task of producing common names; priming the alternative-to-common name connections (e.g., ‘spectacles’→‘glasses’) should allow reverberating activation flow between synonyms.

The results of the interesting recent study by Ibrahim, Cowell, and Varley (2017) are relevant to our theoretical claims. Ibrahim et al. studied both cross-language translation times in bilinguals and synonym “translation” times in monolinguals, where printed words were presented and participants were required to verbally generate a single synonym of each word (e.g., help→“assistance”, prison→“jail”). They replicated Kroll and Stewart’s (1994) finding of an asymmetry in translation times in bilinguals; for both of two groups of Russian-English bilinguals (who differed in their degree of L2 usage), translating L2-to-L1 was faster than translating L1-to-L2 words (1148 vs. 1240 ms). Ibrahim et al. also found that translating from lower-to-higher frequency words (e.g., spectacles→“glasses”) was faster than translating from higher-to-lower frequency synonyms (e.g., glasses→“spectacles”), and by a considerable degree (1702 vs. 2842 ms). At first glance these results are contrary to our proposal that there are *stronger* functional connections from L1-to-L2 and from common-to-rarer synonymous names than in the reverse directions, as this might expect slower (and not faster) L2-to-L1 and lower-to-higher frequency translation times. However, a critical aspect of our account is that the lexical connections we propose are between the representations responsible for spoken word production. In Ibrahim et al.’s experiments, the printed stimulus words were never spoken; rather they had to access the semantic system (and we assume that the process generating synonyms is unlikely to rely solely on lexical inter-connections). In our experiments, the printed

words were distractors in the task of object naming, which requires output from the semantic system to generate speech.

In synthesis, we propose that the phonological lexical representations responsible for speech production have functionally stronger L1-to-L2 and common-to-rarer name connections than in the reverse directions. However, we grant that the orthographic lexical representations responsible for access to semantics may have stronger L2-to-L1 and rarer-to-common name connections rather than the other way round (as Kroll & Stewart’s revised hierarchical model proposes). We would like to stress that the motivation for our hypothesis was to account for our findings of the effects of printed distractor words on object naming times. We found that L1 translation distractor words had a larger facilitation effect on naming in L2 than L2 words had L1 naming in L1, which is the opposite pattern to what might be expected by the revised hierarchical model’s claim that L1-to-L2 connections are weaker than L2-to-L1 connections. We also found that more common synonym distractor words had a greater facilitation effect on producing alternative names than less-common synonyms had on producing common object names. It is these similar patterns of results that our theoretical proposals were advanced to explain.

Ibrahim et al. (2017) interpret their results from both monolinguals and bilinguals in terms a common theoretical framework, namely that lexical accessibility is determined by word frequency. We also attempt a common, integrated theoretical framework to interpret our bilingual and monolingual results, but have arrived at an apparently opposite conclusion concerning the pattern of asymmetries between inter-connected lexical representations responsible for speech production in naming, compared to those required for semantic access.

8.5. Conclusion

We have shown impressive similarities between the facilitation effects of translation and synonym distractors in the picture-word task. Of course, synonyms cannot be the same as translation equivalents in all contexts; translations, unlike synonyms, are not communicatively interchangeable (Costa, 2005), and synonyms do not always produce the same results as translation equivalents in comprehension or memory tasks, where they do not have the “same cognitive status” (Francis, 2005). It is clear that not all our word pairs judged as (and rated to be) synonyms actually have exactly the same meanings (e.g., crocodile and alligator refer to distinguishably different classes of objects, but were rated highly as being closely synonymous). There is socio-linguistic pressure to differentiate synonyms (Clark, 1987), except perhaps for persisting dialect equivalents (e.g., jug–pitcher and biscuit–cookie in British- vs. American-English). The synonym pairs we used as alternative names (and as related distractors) were fully accepted by our participants as being plausible target names for our pictured objects. With these stimuli, we have demonstrated that synonyms for monolinguals behave very similarly to translations for bilinguals in the picture-word task. This suggests that there are important commonalities in the underlying mechanisms of lexical selection in bilinguals and monolinguals. In all speakers, a number of candidate words will be activated by the semantic representation necessary to express an intended idea in speech or to name an object in the laboratory. Which word will be selected will depend upon the levels of activation of lexical representations from various sources but most notably from a language control system that is sensitive to language choice, task requirement, and more generally the speaker’s judgements concerning appropriateness for the communicative context. How speakers manage to find the right words to say is indeed a hard problem, but it will be better understood by the study of commonalities between bilingual and monolingual lexical selection.

Appendix A

The stimuli used in Experiment 1.

Object	Spanish distractors		English distractors	
	Name	Control	Translation	Control
BED	cama	dado	bed	air
BOOK	libro	puerta	book	hair
BUTTERFLY	mariposa	persiana	butterfly	bodyguard
CAR	coche	espejo	car	boy
CAT	gato	dedo	cat	bus
CHAIR	silla	barco	chair	cream
COW	vaca	gorra	cow	pot
DOG	perro	hombro	dog	gun
EAR	oreja	avión	ear	map
FORK	tenedor	carpeta	fork	deer
HORSE	caballo	cerebro	horse	dress
NOSE	nariz	vaso	nose	tape
OWL	búho	sello	owl	lid
PENCIL	lápiz	peine	pencil	kidney
PUMPKIN	calabaza	libélula	pumpkin	luggage
SCISSORS	tijeras	bufanda	scissors	mattress
SHIRT	camisa	pierna	shirt	knife
SOCK	calcetín	pimiento	sock	cart
SQUIRREL	ardilla	cinturón	squirrel	sidewalk
SWAN	cisne	fresa	swan	bean
TABLE	mesa	ojo	table	paper
TIE	corbata	ascensor	tie	pig

Appendix B

The stimuli used in Experiment 2.

Object	Swedish distractors		English distractors	
	Name	Control	Translation	Control
ARROW	pil	skål	arrow	table
BARREL	tunna	paraply	barrel	donkey
BED	säng	pil	bed	fox
BIRD	fågel	tårta	bird	key
BOTTLE	flaska	stege	bottle	eye
BOW	rosett	groda	bow	pig
BOWL	skål	fågel	bowl	rabbit
BUTTERFLY	fjäril	säng	butterfly	envelope
BUTTON	knapp	svamp	button	lemon
CAKE	tårta	knapp	cake	barrel
CAR	bil	gunga	car	squirrel
CARROT	morot	rosett	carrot	well
CHAIR	stol	kanin	chair	dress
CLOUD	moln	gaffel	cloud	onion
DONKEY	åsna	blomma	donkey	fence
DRESS	klänning	brunn	dress	turtle
ENVELOPE	kuvert	lök	envelope	mushroom
EYE	öga	fjäril	eye	bird
FENCE	staket	gris	fence	glove
FLOWER	blomma	slips	flower	chair
FORK	gaffel	bil	fork	tie
FOX	räv	handske	fox	carrot

FROG	groda	citron	frog	cake
GLOVE	handske	staket	glove	strawberry
KEY	nyckel	sköldpadda	key	bed
LADDER	stege	kuvert	ladder	bow
LEMON	citron	fönster	lemon	cloud
MUSHROOM	svamp	tunna	mushroom	window
ONION	lök	klänning	onion	snake
PIG	gris	jordgubbe	pig	ladder
PINEAPPLE	ananas	öga	pineapple	bottle
RABBIT	kanin	bord	rabbit	bowl
SNAKE	orm	moln	snake	umbrella
SQUIRREL	ekorre	nyckel	squirrel	flower
STRAWBERRY	jordgubbe	stol	strawberry	car
SWING	gunga	flaska	swing	pineapple
TABLE	bord	åsa	table	frog
TIE	slips	ekorre	tie	butterfly
TURTLE	sköldpadda	morot	turtle	button
UMBRELLA	paraply	räv	umbrella	swing
WELL	brunn	ananas	well	fork
WINDOW	fönster	orm	window	arrow

Appendix C

The stimuli used in Experiment 3.

Object	English distractors		French distractors	
	Name	Control	Translation	Control
APPLE	apple	skirt	pomme	jupe
ARM	arm	dog	bras	chien
BARREL	barrel	trousers	tonneau	pantalon
BASKET	basket	wheel	panier	roué
BED	bed	fish	lit	poisson
BELL	bell	fence	cloche	cloture
BOX	box	hand	boite	main
CAR	car	glass	voiture	verre
COW	cow	thumb	vache	pouce
DESK	desk	foot	bureau	pied
DOG	dog	barrel	chien	tonneau
DOOR	door	bed	porte	lit
FENCE	fence	tie	cloture	cravat
FISH	fish	window	poisson	fenetre
FOOT	foot	house	pied	maison
GLASS	glass	box	verre	boite
HAND	hand	car	main	voiture
HEART	heart	door	coeur	porte
HOUSE	house	arm	maison	bras
LEMON	lemon	pencil	citron	crayon
PENCIL	pencil	tree	crayon	arbre
SHEEP	sheep	swing	mouton	balancoire
SKIRT	skirt	cow	jupe	vache
SWING	swing	sheep	balancoire	mouton
THUMB	thumb	bell	pouce	cloche
TIE	tie	lemon	cravate	citron
TREE	tree	desk	arbre	bureau
TROUSERS	trousers	apple	pantalon	pomme
WHEEL	wheel	basket	roué	panier
WINDOW	window	heart	fenetre	coeur

Appendix D

The stimuli used in Experiment 4.

Object	Distractors			
	Name	Control	Synonym	Control
AXE	axe	rye	hatchet	seafood
BARREL	barrel	tissue	keg	pub
BULLET	bullet	finger	ammunition	institution
CAR	car	boy	automobile	videotape
CLOCK	clock	guard	timepiece	birdhouse
CROCODILE	crocodile	blueberry	alligator	spaceship
DOG	dog	eye	hound	torch
DONKEY	donkey	fabric	mule	cord
DRESS	dress	train	frock	shrew
FROG	frog	pill	toad	vest
GLASSES	glasses	stomach	spectacles	cellophane
GLOBE	globe	strap	earth	paper
GORILLA	gorilla	stadium	ape	rug
GUN	gun	bed	pistol	spider
JUMPER	jumper	meadow	sweater	dessert
LORRY	lorry	mango	truck	field
PIG	pig	oil	swine	crate
SNAKE	snake	trash	serpent	orchard
SOFA	sofa	tomb	couch	apple
SPADE	spade	flute	shovel	poster
WIZARD	wizard	pillow	sorcerer	lollipop

Appendix E

The stimuli used in Experiment 5.

Object	Distractors			
	Name	Control	Synonym	Control
AXE	axe	glasses	hatchet	serpent
BARREL	barrel	feather	cask	pistol
BUCKET	bucket	ghost	pail	tomb
CAR	car	wizard	automobile	plume
CASTLE	castle	pig	fort	pail
CHICKEN	chicken	grave	hen	sorcerer
DOG	dog	car	hound	cask
DRESS	dress	gun	frock	hen
FEATHER	feather	dog	plume	spectacles
GHOST	ghost	axe	phantom	hatchet
GLASSES	glasses	snake	spectacles	automobile
GRAVE	grave	chicken	tomb	hound
GUN	gun	bucket	pistol	phantom
PIG	pig	castle	swine	frock
SNAKE	snake	dress	serpent	swine
WIZARD	wizard	barrel	sorcerer	fort

Appendix F

Means of trimmed arithmetic mean naming latencies (in milliseconds) in each condition for Related (Rel.) and Unrelated (Un.) distractors for each naming language or naming task. Facilitation (Facil.) = Unrelated minus Identity or Translation. Also shown, for comparison, are the facilitation effects reported in the tables of mean harmonic mean naming times.

	Naming in L1			Naming in L2			From tables	
	Rel.	Un.	Facil.	Rel.	Un.	Facil.	L1	L2
<i>Experiment 1</i>								
(a) Identity	718	838	120	718	824	106	100	94
(b) Translation	815	820	5	814	848	34	5	38
<i>Experiment 2</i>								
(a) Identity	733	912	179	729	901	172	179	165
(b) Translation	852	903	51	770	917	147	52	128
<i>Experiment 3</i>								
(a) Identity	760	879	119	768	915	147	110	134
(b) Translation	812	852	40	838	918	77	42	76
	Common naming			Alternative naming			From tables	
	Rel.	Un.	Facil.	Rel.	Un.	Facil.	Com.	Alt.
<i>Experiment 4</i>								
(a) Identity	821	893	72	850	944	94	69	93
(b) Synonym	901	906	5	903	934	31	4	35
<i>Experiment 5</i>								
(a) Identity	770	871	101	842	978	136	85	114
(b) Synonym	848	864	16	920	974	54	12	51

The percentage of naming responses additionally excluded (over those quoted in the text) because they exceeded the 2.5 standard deviation threshold in each Experiment were: Experiment 1 = 2.81%; Experiment 2 = 2.59%; Experiment 3 = 2.60%; Experiment 4 = 2.77%; and Experiment 5 = 2.21%.

Appendix G. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.cognition.2017.09.014>.

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